IEEE P3335 Standard

IEEE-3335 Working Group

Contents

1. Overview (Informative)

This document defines the architectural framework, performance expectations, and interoperability objectives for **TimeCard** devices; modular timing subsystems that provide standardized, high precision time, phase, and frequency services to a host system.

The primary purpose of this specification is to establish a consistent structural and logical framework that accommodates various hardware implementation approaches (e.g., PCIe add-in cards, deeply embedded silicon IP, or external modules) while facilitating broad industry interoperability. This interoperability allows diverse TimeCard implementations to be seamlessly swappable with minimal to no changes required from the host system, empowering the scalable deployment of precision timing in hyperscale computing, telecommunications, and distributed Artificial Intelligence (AI) infrastructure.

As an introductory and informative chapter, the concepts presented herein provide context for the standard and do not contain normative conformance requirements.

1.1 Purpose and Scope

The **IEEE P3335 TimeCard Specification** establishes the structural, electrical, software, and performance characteristics of TimeCard-based architectures. It describes how hardware subsystems acquire external time references, maintain temporal stability during reference loss (holdover), and accurately distribute disciplined phase and frequency outputs to computing hosts and downstream networks.

The scope of this standardization effort encompasses: - Hardware and software architecture definitions, including unified timescale generation and holdover behavior. - Receive (inbound) and Providing (outbound) timing interface behaviors (e.g., GNSS, PTP, 1PPS, and PCIe PTM). - Out-of-band and in-band management, telemetry, and security mechanisms (e.g., I²C, IPMI, REST). - Quantifiable performance metrics (e.g., ADEV, TDEV, MTIE) and environmental operating constraints. - Structural methodologies for testing and claiming conformance, including functional, performance, and environmental validation procedures based on traceable metrology.

This document is designed to apply equally across: - Vendor-specific commercial mass-production implementations. - Open-source hardware reference designs. - Laboratory, exploratory, or research-grade prototypes.

By defining a **standardized timing subsystem interface**, the specification establishes a foundation where different TimeCards from competing vendors can be easily interchanged, upgraded across generations, or co-deployed across diverse host platforms while maintaining stringent functional and timing compatibility.

1.2 Motivation

Modern computing workloads increasingly depend on highly precise and deterministic hardware timing. Modern applications—such as **AI model training clusters, high-frequency financial trading, globally distributed Spanner-style databases, and 5G/6G cellular networks**—frequently require sub-microsecond or nanosecond-class synchronization across thousands of independent nodes.

Historically, achieving this level of timing relied on bespoke, vendor-specific implementations or tightly coupled architectures that lacked modularity and scale. The **TimeCard architecture** addresses these historical bottlenecks by introducing: - A standardized hardware abstraction layer and signal footprint (e.g., standardized 1PPS, 10 MHz, and Time of Day bounds).
- A unified management and telemetry framework supporting diverse transports for configuration and observability.
- A common timing distribution mechanism capable of bypassing host operating system latencies using hardware-based timestamping (e.g., PCIe Precision Time Measurement or hardware PTP).
- A clear, falsifiable path for cross-vendor conformance certification, grounded in traceable metrology and repeatable test procedures.

The ultimate motivation is to foster an open, competitive ecosystem for precision time distribution.

1.3 Design Philosophy

The P3335 TimeCard specification is founded on several core engineering principles:

Principle	Description
Modularity	The TimeCard operates as a self-contained subsystem that fully abstracts the complexities of oscillator disciplining away from the host CPU.
Interoperability	Management and timing interfaces are standardized to facilitate plug-and-play architectural compatibility across different server fleets and boundary clocks.
Scalability	The architecture gracefully scales from single-node isolated edge servers up to globally synchronized hyperscale network domains relying on unified ensemble clocks.
Determinism	Hardware-assisted generation and timestamping provide highly predictable, low-jitter, and low-latency outputs, immune to software scheduling variations.
Traceability	Subsystem logic supports establishing an unbroken chain of measurements reflecting recognized international standards (e.g., UTC, TAI), measured using calibrated metrology equipment.
Security and Integrity	Remote management channels and firmware provisioning protocols rely on modern authenticated and cryptographically verifiable mechanisms, including provisions for data sanitization.

1.4 Relationship to Other Standards

This specification aligns with and periodically references multiple established global timing and measurement standards. Key related standards include:

- **IEEE Std 1588™-2019 (PTPv2.1):** Precision Time Protocol for generalized network synchronization.
- **IEEE Std 802.1AS™-2020:** Timing and Synchronization for Time-Sensitive Applications.
- **ITU-T G.810 / G.8260 / G.8271:** Definitions and boundary parameters for carrier-grade synchronization networks.
- **IEEE Std 1139™ / IEEE Std 1193™:** Standard definitions and environmental testing methodologies for fundamental frequency metrology quantities (ADEV, TDEV, MTIE).
- **PCI-SIG PCIe Base Specification:** Specifically outlining Precision Time Measurement (PTM) for in-band hardware timestamping.
- **Open Compute Project (OCP) Time Appliances Project (TAP):** The foundational open-hardware consortium that incubated the original TimeCard concept.

Where applicable, this document cites these standards normatively to maintain rigid technical consistency across the industry.

1.5 Structure of the Specification

The standard is organized sequentially into the following major clauses:

Chapter	Description
1. Overview	Provides the motivation, high-level scope, and conceptual structure (Informative).
2. Normative References	Lists the foundational standards essential for implementing this specification.
3. Definitions, Acronyms and Abbreviations	Clarifies exact terminology and abbreviations utilized throughout the text.
4. Conformance	Outlines the formal terminology and methodology for claiming implementation compliance.
5. Architecture	Defines the core TimeCard blueprint, oscillator interactions, and boundary definitions.
6. Performance Specifications	Details the mathematical metrics and methodologies used to quantify stability and accuracy.
7. Timing Interfaces	Specifies the electrical and logical conduits for receiving and providing synchronization.
8. Control Interfaces	Defines the data structures required for out-of-band and in-band management.
9. Environment	Establishes baselines for physical survivability, thermal tolerance, and RF emissions.
10. Applications and Best Practices	Offers structural guidance on real-world deployment and operational optimization (Informative).
Annex A. Metrics	Provides definitions and measurement methodologies for timing and frequency metrics (Informative).
Annex B. Test Procedures	Provides standardized procedures for functional, performance, and environmental testing (Informative).
Annex C. Bibliography	Lists informational references and supplementary reading.

1.6 Intended Audience

This specification is drafted distinctly for: - Hardware, software, and systems engineers developing TimeCard-compatible commercial products or reference designs.
- Datacenter architects and system integrators designing synchronized infrastructure and planning for scale-out timing topologies.
- Network operators and site-reliability engineers managing latency-sensitive distributed environments who require granular telemetry.
- Scientists and researchers requiring high-fidelity timestamping for metrology and specialized physics experimentation. - Test and validation engineers utilizing the defined procedures to empirically grade hardware performance and claim conformance. - Affiliated standards bodies (e.g., ITU-T or OCP) seeking structural harmonization with modern end-system timing frameworks.

Readers are expected to possess a foundational understanding of frequency control, phase-locked loops, phase noise characteristics, and packet-based synchronization (e.g., PTP, NTP).

1.7 Strategic Goals

The widespread adoption of the TimeCard architecture aims to achieve the following outcomes: - **Broad vendor-neutral interoperability** among specialized timing payloads and generic host computers, enabling a mature, commercial off-the-shelf (COTS) ecosystem.
- **Drastically improved absolute precision** within distributed databases, cellular networks, and distributed AI compute clusters by driving the timescale directly to the hardware bus (e.g., PCIe).

- **Lowered integration friction and cost** resulting from strictly defined hardware abstraction interfaces and deterministic holdover behaviors.
- **Enhanced operational observability** and traceability to support stringent financial, telecommunications, and governmental regulatory frameworks.
- **Consistent metrology evaluations** resulting from standardized, falsifiable performance testing and conformance guidelines.

1.8 Document Status and Origin

This architectural framework heavily draws its foundations from the pioneering work accomplished within the **Open Compute Project (OCP) Time Appliances Project (TAP)**. The transition of this architecture into the formal **IEEE P3335** standardization process reflects the industry's demand for a rigorously governed, universally recognized timing hardware standard.

Contributors, silicon vendors, and system integrators are actively encouraged to submit empirical performance data, interoperability reports, and technical proposals to inform the ongoing evolution of this specification.

2. Normative References

The following documents are referred to in the text in such a way that some or all of their content constitutes requirements of this document. For dated references, only the edition cited applies. For undated references, the latest edition of the referenced document (including any amendments or corrigenda) applies.

Note: In IEEE drafting style, this section only contains references that are indispensable for the application of the document (normative). References that are supplementary, educational, or optional are located in the Informative Bibliography (see Section 2.2).

2.1 Normative References

- **IEEE Std 1139™**, *IEEE Standard Definitions of Physical Quantities for Fundamental Frequency and Time Metrology.*
- **IEEE Std 1193™**, *IEEE Guide for Measurement of Environmental Sensitivities of Standard Frequency Generators.*
- **IEEE Std 1588™-2019**, *IEEE Standard for a Precision Clock Synchronization Protocol for Networked Measurement and Control Systems* (Precision Time Protocol / PTPv2.1).
- **IEEE Std 802.1AS™-2020**, *IEEE Standard for Local and Metropolitan Area Networks—Timing and Synchronization for Time-Sensitive Applications.*
- **IETF RFC 5905**, *Network Time Protocol Version 4: Protocol and Algorithms Specification.*
- **ITU-T Recommendation G.810**, *Definitions and terminology for synchronization networks.*
- **ITU-T Recommendation G.8260**, *Definitions and terminology for synchronization in packet networks.*
- **ITU-T Recommendation G.8271**, *Time and phase synchronization aspects of telecommunication networks.*
- **PCI-SIG PCIe Base Specification 5.0** (or later), specifically citing the *Precision Time Measurement (PTM)* subclauses.
- **OCP Time Appliances Project (TAP)**, *TimeCard Specification*, Open Compute Project Foundation.

2.2 Bibliography (Informative References)

The bibliography lists documents that are referenced for informative, educational, or background purposes within the text. Their implementation is not required to claim conformance to this standard.

Note: In the final compiled IEEE document, this section will likely be moved to the final Annex (e.g., Annex A: Bibliography).

- **CERN / GSI White Rabbit**, *White Rabbit Specification and Documentation* (Sub-nanosecond Ethernet synchronization).
- **IEC 61000-6-x Series**, *Electromagnetic compatibility (EMC) - Generic standards limit for emission and immunity.*
- **IEEE Communications Surveys & Tutorials**, *Multi-constellation GNSS receiver performance studies.*
- **ISO/IEC 17025**, *General requirements for the competence of testing and calibration laboratories.*
- **ITSF Proceedings**, *International Telecom Sync Forum (ITSF) - Industrial case studies and best practices.*
- **NIST Special Publication 1065**, W.J. Riley, *Handbook of Frequency Stability Analysis* (2008).
- **NIST Technical Note 1952**, *Introduction to PTP and synchronization techniques.*
- **NTP.org**, *NTP Reference Implementation (ntpd) open-source software project.*
- **OCP Github Repository**, *Open Compute Project TimeCard Hardware and Firmware Repositories* (Open-source schematics and HDL).
- **OCP TAP Interoperability Whitepaper**, *Framework for multi-vendor validation and testing definitions.*
- **OCP TAP Wiki**, *Community resources, meeting notes, and specification archives.*
- **Turba Technologies**, *Turba Analytics Documentation - AI workload and distributed timing analytics platform.*

End of Chapter – Normative References

3. Definitions, Acronyms, and Abbreviations

This chapter defines the specialized terms, acronyms, and abbreviations utilized throughout the IEEE P3335 specification.

3.1 Definitions

For the purposes of this document, the following terms and definitions apply. The *IEEE Standards Dictionary Online* should be consulted for terms not defined in this clause.

- **Accuracy:** The degree of conformance or closeness of a measured time or frequency value to a defined primary reference (e.g., UTC).
- **Asymmetry Calibration:** The process of calculating and correcting unequal signal propagation delays in bi-directional timing links.
- **Control Plane:** The logical communication path dedicated to configuration, telemetry, and operations management rather than active time distribution.
- **Data Plane:** The logical or physical communication path strictly responsible for carrying phase, frequency, and time signals to or from the host or network.
- **Data Plane Latency:** The deterministic or variable propagation delay between a time source and its destination hardware.
- **Deterministic Behavior:** A system characteristic where the timing response under specified operational conditions is highly predictable and repeatable.
- **Disciplining:** The continuous process of steering a local oscillator's frequency and phase to align with a superior external reference signal.

- **Ensemble Clock:** A highly stable composite time source created by mathematically combining multiple independent references via weighting or consensus algorithms.
- **Environmental Chamber:** A specialized testing enclosure used to strictly control ambient temperature, humidity, and atmospheric conditions for hardware thermal testing.
- **Frequency Counter:** A measurement instrument used to determine the exact signal frequency or period over a defined averaging interval.
- **Granularity:** The minimum distinguishable or settable unit of change in a time-domain or frequency-domain output.
- **Hitless Switching:** The ability of a timing device to execute a reference failover or changeover without inducing a measurable phase discontinuity to the downstream consumers.
- **Holdover:** A mode of operation where a timing subsystem continues to maintain accurate time based solely on the historical stability of its internal oscillator after losing its external reference.
- **Holdover Error:** The progressively increasing deviation or drift of the subsystem's timescale from an ideal reference while operating in a holdover state.
- **Host System:** The overarching computing, telecommunications, or networking platform that integrates a TimeCard through a standardized interface boundary.
- **Loop Bandwidth:** The effective frequency range over which a disciplining Phase-Locked Loop (PLL) tracks the reference; determining how fast the loop responds to changes.
- **Lock Time:** The duration required for the subsystem to acquire a reference and assert phase/frequency lock after a cold startup or reference transition.
- **Management Interface:** The designated control channel (e.g., SMBus, IPMI, REST) used strictly for configuration, diagnostics, and firmware management.
- **Oscilloscope (TIE Mode):** Measurement equipment configured to capture the Time Interval Error (TIE) of repetitive high-frequency signals.
- **Phase Alignment:** The absolute difference in timing between the corresponding edges of two separate signals; typically measured in nanoseconds or picoseconds.
- **Power Analyzer:** An instrument deployed to measure precise current draw, power sequencing, and consumption events over time.
- **Precision:** The statistical repeatability, resolution, or variance of a measurement under identical operating conditions.
- **Primary Reference Clock:** The highest-stability, autonomous time or frequency source in a synchronization hierarchy (as defined by ITU-T G.811).
- **Providing Interface:** The outbound physical or logical mechanism through which the TimeCard distributes precise time, phase, and frequency representations to the host or downstream devices.
- **Receive Interface:** The inbound physical or logical mechanism through which the TimeCard acquires an external reference representation.
- **Reference Signal:** Any external, traceable timing input actively utilized to discipline the local oscillator.
- **Resolution:** The absolute smallest incremental step, quantization, or granularity of a time or frequency measurement that the system can distinguish.
- **Rubidium Oscillator:** A highly stable atomic oscillator utilizing rubidium-87 vapor transitions to provide exceptional long-term frequency stability.
- **TimeCard:** A modular, hardware-based timing subsystem that abstracts synchronization complexity to deliver standardized phase, frequency, and time services to a host system.
- **Time Interval Counter (TIC):** A highly precise laboratory instrument designed to measure time differences between distinct electrical signal edges with picosecond-level resolution.
- **Time Jitter:** The short-term variation or instability of a time-domain signal (such as a 1PPS edge) from its ideal position, frequently reported as an RMS or peak-to-peak value.
- **Traceability:** A documented, unbroken mathematical chain of calibrations linking a local measurement back to recognized primary international standards (e.g., UTC).
- **Unified Timescale:** A single internal temporal scale actively maintained by the TimeCard, from which all separate outbound interfaces derive coherent, phase-aligned realizations.

3.2 Acronyms and Abbreviations

- **1PPS:** One Pulse Per Second
- **ADEV:** Allan Deviation
- **ASIC:** Application Specific Integrated Circuit
- **BMC:** Baseboard Management Controller
- **CXL:** Compute Express Link
- **CSAC:** Chip-Scale Atomic Clock
- **DDS:** Direct Digital Synthesizer
- **EMC:** Electromagnetic Compatibility
- **EMI:** Electromagnetic Interference
- **ESD:** Electrostatic Discharge
- **FPGA:** Field Programmable Gate Array
- **GNSS:** Global Navigation Satellite System (e.g., GPS, Galileo, GLONASS)
- **gRPC:** Google Remote Procedure Call
- **I2C:** Inter-Integrated Circuit
- **I3C:** Improved Inter-Integrated Circuit
- **IPMI:** Intelligent Platform Management Interface
- **IRIG:** Inter-Range Instrumentation Group
- **MAC:** Media Access Control
- **MMIO:** Memory-Mapped Input/Output
- **MTBF:** Mean Time Between Failures
- **MTIE:** Maximum Time Interval Error
- **NC-SI:** Network Controller Sideband Interface
- **NTP:** Network Time Protocol
- **OCP TAP:** Open Compute Project Time Appliances Project
- **OCXO:** Oven-Controlled Crystal Oscillator
- **PCIe:** Peripheral Component Interconnect Express
- **PHY:** Physical Layer
- **PLL:** Phase-Locked Loop
- **PN:** Phase Noise
- **PTM:** Precision Time Measurement
- **PTP:** Precision Time Protocol
- **REST:** Representational State Transfer
- **RoHS:** Restriction of Hazardous Substances
- **SCPI:** Standard Commands for Programmable Instruments
- **SEC:** Securities and Exchange Commission
- **SMA:** SubMiniature version A (connector)
- **SMBus:** System Management Bus
- **SNMP:** Simple Network Management Protocol
- **SWaP-C:** Size, Weight, Power, and Cost
- **TAI:** International Atomic Time
- **TCXO:** Temperature-Compensated Crystal Oscillator
- **TDEV:** Time Deviation
- **TIC:** Time Interval Counter
- **TIE:** Time Interval Error
- **TPM:** Trusted Platform Module
- **ToD:** Time of Day
- **USB:** Universal Serial Bus
- **UTC:** Coordinated Universal Time
- **WiWi:** Wireless two-Way interferometry
- **WR:** White Rabbit
- **WWVB:** Radio Station WWVB (NIST)

End of Chapter – Definitions, Acronyms, and Abbreviations

> **Editor's Note (Rodney C., P3335 Working Group):** This document currently serves as a structural guideline for defining what will eventually go into the Conformance clause of the P3335 standard. The intention is to align the working group on how to draft normative requirements.

4. Conformance Guidelines & Methodology

4.1 IEEE-SA Terminology Standards

IEEE-SA strictly regulates the terminology used to express requirements within a standard. The following boilerplate text is provided by the IEEE-SA template and cannot be edited by the Working Group.

Requirements placed upon conformant implementations of this standard are expressed using the following specific keywords:

1. **Shall:** Used exclusively to indicate **mandatory requirements** (must-have features to claim conformance).
2. **May:** Used exclusively to describe **implementation or administrative choices** ("may" means "is permitted to"). *Note that "may" and "may not" mean precisely the same thing in this context.*
3. **Should:** Used exclusively for **recommended choices**. The behaviors described by "should" and "should not" are both permissible, but one is clearly more desirable than the other.

4.1.1 Prohibited and Deprecated Terminology

- **Must:** IEEE-SA strictly prohibits the use of "must" to describe requirements because it is frequently confused with "shall". The P3335 text **cannot** use "must." (If an unavoidable physical limitation exists, "must" may sometimes be justified, but it is generally deprecated in modern drafting).
- **Will / Can:** These words are used only for statements of fact or capability, completely devoid of conformance implications.

These three core keywords (*shall, should, may*) are critical. They instruct an engineer exactly how to implement the standard. Without the strategic use of these words, the document functions merely as a "Recommended Practice" rather than a true Standard.

4.2 Structural Methodologies for Conformance

Within IEEE standards, there are three primary methodologies for organizing and expressing these requirements.

4.2.1 Methodology A: Conformance Clause Only

In this approach, normative clauses (Clause 4 and up) and normative annexes avoid using the three keywords entirely, instead using descriptive verbs like "is" and "are" to explain what an implementer does. A single, dedicated Conformance Clause at the beginning uses the keywords by referencing the subsequent normative clauses (e.g., *"An implementation of this standard shall implement only one of the following: a) Clause 4, b) Clause 6, c) Subclause 8.2"*). * **Example:** IEEE 802.1

Advantages: * The Conformance Clause serves as a "one-stop shop" to quickly understand the core requirements. * It easily supports a hierarchical tree of requirements and options (e.g., *"If you implement high-level option A, you shall implement subclause 4.5"*).

Disadvantages: * Standards authors frequently slip and use "is" for concepts unrelated to conformance, which causes ambiguity. * Authors inevitably accidentally use the three keywords within the normative text anyway, creating contradictions with the central Conformance Clause.

4.2.2 Methodology B: Interspersed Without Conformance Clause

In this approach, the normative clauses and annexes directly use the three keywords (*shall, should, may*). Words like "is" and "can" are reserved solely for factual statements with no conformance implication. There is no central Conformance Clause; instead, the requirement tree is reflected purely through the structural organization of the document (e.g., Subclause 6.8.3 might be explicitly titled *"Feature foobar (Optional)"*). * **Example:** IEEE 1588

Advantages: * Avoids the ambiguities and accidental contradictions of Methodology A. * Conformance is immediately clear as the implementer reads through the technical text.

Disadvantages: * Places a heavier burden on the implementer to hunt through the entire document to find all relevant "shalls." * It is significantly more difficult for authors to document a complex, mutually exclusive tree of dependencies.

4.2.3 Methodology C: Interspersed With Conformance Clause

This is a hybrid approach. It uses the dispersed keywords of Methodology B but adds a high-level Conformance Clause to summarize the major requirement trees.

The Conformance Clause provides the high-level roadmap (e.g., *"If you implement high-level option A, you shall implement subclause 4.5"*). Within subclause 4.5 itself, the text will state things like, *"The port shall transmit…"* which implicitly means *"If you support 4.5, you shall do this."*

4.3 P3335 Working Group Recommendations

4.3.1 Recommended Drafting Methodology

> **Recommendation (Rodney C.):** Proceed with drafting the subsequent text using the **Interspersed Technique** (Methodology B).

As the P3335 project matures, the Working Group can evaluate whether a high-level Conformance Clause (upgrading to Methodology C) is necessary. Because P3335 is likely to develop a complex feature tree, a summarizing Conformance Clause will likely prove highly beneficial in later drafts.

4.3.2 Protocol Implementation Conformance Statement (PICS)

> **Recommendation (Rodney C.):** Strongly avoid creating a PICS annex.

A PICS typically takes all the dispersed requirements and repeats them in a massive hierarchical table. Some engineers prefer reading tables over text for compliance checking, which is why PICS annexes exist in many IEEE standards.

The critical disadvantage: A PICS *will* inevitably fall out of sync with the main text of the standard during drafting and iterative revisions. This creates massive contradictions for implementers. Adding disclaimers like *"When there is a contradiction, ignore the PICS"* is often insufficient. *(Note: IEEE 802.1Q heavily suffers from this exact PICS desynchronization problem).*

TimeCard Architecture Specification

The establishment of a standard architecture for TimeCards plays a critical role in enabling interoperability among diverse implementations. By defining a consistent framework, different vendors can design and manufacture TimeCards with varying capabilities, performance levels, and core technologies, while maintaining full plug-and-play compatibility with any compliant host. This standardization fosters an open ecosystem, simplifies hardware integration, and enables seamless substitution or generational upgrades of TimeCards without requiring significant host system redesign.

5.1 - Architecture Overview

A **TimeCard** is a modular subsystem designed to interface with a computing host system through a standardized hardware and software interface. Its primary purpose is to deliver a stable, accurate, deterministic, and reliable source of time (in the form of time of day, phase, frequency, or any combination thereof) to the host system.

5.1.1 Rationale for Dedicated Timing Subsystems

Modern host systems (such as high-performance servers, edge compute nodes, and telecommunications routers) typically lack the internal capabilities required to maintain sub-microsecond or nanosecond-class synchronization. Host limitations generally include unpredictable software and operating system scheduling latencies that interfere with precise clock steering, large thermal profiles that degrade standard quartz oscillators, and a lack of specialized hardware for bounded-latency time transfer or hardware timestamping.

The TimeCard overcomes these host limitations by completely offloading critical timing functions—such as phase-locked loops (PLLs), holdover tracking, and signal timestamping—to a dedicated, physically isolated subsystem. By incorporating a TimeCard, a host system gains enhanced, deterministic timekeeping and synchronization capabilities without requiring a fundamental redesign of the host's primary processing architecture.

5.1.2 Implementation Modularity

While a common physical manifestation of a TimeCard is a discrete add-in card (such as a PCI Express card) inserted into a server chassis, a TimeCard system is fundamentally defined by its logical interfaces and behaviors rather than its physical constraints.

Alternate valid implementations include, but are not limited to: * A dedicated intellectual property (IP) block directly embedded into a System-on-Chip (SoC) or integrated onto a server motherboard. * An external desktop or ruggedized module temporarily or permanently connected to the host system via a hot-pluggable or peripheral interface (e.g., USB, Thunderbolt).

Any subsystem that meets the architectural boundaries and interface definitions defined within this standard is classified as a TimeCard for the purposes of conformance.

5.1.3 Hardware Timestamping

To preserve the integrity and determinism of timing, it is strongly recommended that both the providing and receiving interfaces implement **hardware-based timestamping**. Hardware timestamping enables timing information to be generated and measured directly within hardware logic, avoiding non-deterministic delays caused by software stacks, interrupt latencies, and/or operating system scheduling.

Hardware timestamping can be realized through dedicated physical signals, such as a **Pulse-Per-Second (PPS)** output, or through advanced in-bus implementations, such as **Precision Time Measurement (PTM)** within modern **PCIe** architectures. By driving the clock boundaries directly into the hardware bus, these mechanisms enable low-latency, deterministic time delivery. This allows distributed databases and cellular packet schedulers to achieve sub-microsecond absolute precision, improving cross-vendor interoperability among TimeCard and host designs.

5.2 - Core Timing Architecture

At its core, every TimeCard is built around at least one **frequency source** (with quantified stability), which serves as the foundational source of precise timing. This source oscillator function is complemented by one or more interfaces that enable the TimeCard to **receive** and/or **generate** and **distribute** time of day, phase, and frequency information to and from the host system.

5.3 - Inbound (Receiving) Signal Interface

The **receive interface** provides a means for the TimeCard to synchronize its oscillator function to an external reference. Depending on the deployment environment and the required accuracy, this interface may take multiple forms. Common examples include **Global Navigation Satellite System (GNSS)** receivers (e.g., GPS, Galileo, GLONASS, BeiDou), or other precision synchronization methods such as **Precision Time Protocol (PTP)**, **Network Time Protocol (NTP)**, **White Rabbit (WR)**, **WiWi**, **WWVB**, or **Pulse-Per-Second (PPS)** inputs. These interfaces allow the TimeCard to discipline its oscillator function and maintain alignment with an external time source. The external references may or may not be more stable or of lower noise than this oscillator function.

In some configurations, a TimeCard may operate without any inbound external timing input. In this mode, the TimeCard functions in **holdover**, relying solely on the stability of its internal oscillator function to maintain accurate time over a defined interval. Such configurations are particularly useful in environments where external timing references are unavailable, intermittent, or deliberately excluded for security or operational isolation to enable redundancy.

This flexible receive architecture enables TimeCards to support a wide use-case spectrum - from GNSS-disciplined primary time sources at the edge of the network, to deep-indoor boundary clocks relying on PTP *« explain relevance »**, to autonomous isolated holdover systems - while preserving a common and interoperable host interface standard. This ensures that a datacenter can deploy identical host server binaries regardless of the specific external timing source used by the TimeCard.

5.4 - Outbound (Providing) Signal Interface

While the receive interface allows synchronization to an external reference, the **providing interface** ensures that the synchronized time and frequency are accurately distributed to the host system.

A providing interface is a **mandatory component** of every TimeCard. It defines the mechanism by which the TimeCard delivers time of day, phase, frequency, or all, to the host, forming the synchronization channel between TimeCard and host.

Depending on system requirements, the providing interface may consist of a single interface or a combination of multiple concurrent interfaces. Common examples include system bus standards such as **ISA**, **MCA**, **PCI**, and **PCI Express (PCIe)**, as well as peripheral and communication interfaces such as **Serial Bus**, **USB**, **SCSI**, **PCMCIA**, or **LPT**. The selection of interface type directly influences both the data exchange characteristics and the precision of temporal alignment achievable by the host.

5.5 Management and Control Interface (M&CI)

In addition to the inbound and outbound signal interfaces, it is recommended that each TimeCard include at least one **management and control interface**. These interfaces enable configuration, monitoring, diagnostics, firmware management, and status reporting between the TimeCard and the host. A TimeCard without a management interface is acceptable if its operational parameters are fixed or pre-determined, and no runtime monitoring or control is required.

The management interface functions as the **control plane** of the TimeCard, distinct from the **data plane** used for delivering timing and frequency. Through this interface, the host can configure and observe operational parameters such as oscillator state, synchronization source selection, disciplining mode, holdover behavior, temperature compensation, and alarm or fault conditions.

The following M&CI busses are independent of one another, and a TimeCard may utilize more than one kind of bus. Common examples of management and control interfaces include but are not limited to: - **SMBus or I²C** – typically used for low-level configuration and telemetry in embedded environments.

- **IPMI** (Intelligent Platform Management Interface) – for out-of-band management in server-class or rack-scale systems.
- **PCIe Configuration Space Registers** – providing time-related control and status directly over the host bus, including direct memory access to the hardware control registers.
- **Serial or USB interfaces** – enabling firmware updates, diagnostics, or advanced telemetry access.
- **Network-based interfaces** such as REST, gRPC, or SNMP, for distributed or remotely managed timing systems.

To promote interoperability and consistency, all TimeCards **SHOULD** expose a minimum common set of registers and attributes in a standardized format, including (but not limited to): - Current synchronization source and state
- Clock disciplining status
- Phase and frequency offset metrics
- Holdover duration and expected drift
- Alarm and fault indicators
- Firmware version and build metadata

Furthermore, the management interface **SHOULD** support secure firmware update and integrity verification mechanisms to ensure reliability and prevent unauthorized modification. Collectively, these management and control capabilities provide the operational transparency and lifecycle management required for seamless integration of TimeCards into data centers, telecom infrastructure, and AI back-end clusters. In distributed environments, centralized management software can poll these M&CI endpoints to establish a global view of timing health, rapidly identifying degraded oscillators or spoofed GNSS signals before they impact the primary workload.

5.6 - Power, Mechanical, and Environmental Considerations

The subsections apply only if the TimeCard is implemented physically, versus for instance as a firmware function within a larger system. ### 5.6.1 - Power Delivery - TimeCards **SHALL** define and document input power rail voltages and tolerances (e.g., 12 V, 3.3 V).
- If externally powered, protection against reverse polarity **SHALL** be included.
- Deterministic power-up sequencing and optional energy storage for holdover **SHOULD** be supported.

5.6.2 - Mechanical Form Factor

- The physical envelope **SHALL** be documented.
- Acceptable envelope forms include add-in cards (low-profile/full-height), mezzanine, or embedded.
- Mounting **SHALL** withstand insertion/removal; strain relief for all cable ports (electrical or optical) **SHOULD** be included.
- Faceplates **SHOULD** label at least GNSS, PPS, 10 MHz, ToD, and management ports and include indicator LEDs.

5.6.3 - Connectors and I/O

- RF/timing reference signal ports **SHALL** be impedance-matched.
- PPS/10 MHz electrical levels, impedance, and edge polarity (trigger on the rising or falling edge) **SHALL** be specified.
- Data and management ports **SHOULD** have ESD protection and mechanically locking connectors.

5.6.4 - Thermal Design

- Operating temperature range **SHALL** be defined; oscillator drift vs. temperature **SHOULD** be characterized and documented.

- Host airflow assumptions **SHOULD** be documented; heat load during warm-up **SHALL** be specified.

- Over/under-temperature thresholds **SHALL** be reported via a management and control interface, if such an interface is provided.

5.6.5 - Environmental and Reliability

- Shock, vibration, and humidity limits **SHOULD** be stated.

- EMC/ESD compliance **SHOULD** meet target-market standards.

- MTBF and wear-out items **SHOULD** be documented and published.

- Safety, labeling, and disposal requirements **SHOULD** be provided.

5.7 - Reference Signals and Performance Metrics (Normative)

- Methods defined in **Annex A (Metrics)** of this standard, which align with IEEE 1139, IEEE 1193, and ITU-T G.810 / G.8260, **SHOULD** be used for performance-metric definitions and analysis.
- NIST Special Publication 1065 (by W.J. Riley) provides an informative foundation for frequency metrology.
- Requirements **SHOULD** be conditioned on the physical characteristics of the interface type, including but not limited to: electrical balanced or unbalanced signaling, voltage and/or current thresholds, optical fiber classification, and frequency band.
- ITU-T G.703 Clause 19 **MAY** be used as a reference for synchronous signaling.
- Measurement equipment bandwidths in Hertz and trace averaging settings **SHALL** be explicitly documented alongside all reported performance results.

5.7.1 - Unified Timescale (Normative)

A unified timescale comes from a single oscillator function which is published in multiple distribution formats each approximating the ideal of the timescale to the capabilities of each format.

A TimeCard **SHALL** generate a single unified timescale and **SHALL** publish it across all Outbound Interfaces.

All boundaries between adjacent seconds of signals from the the providing interface of the same TimeCard instance **SHALL** align to within a time tolerance that is documented and published.

5.7.2 - Output Signal Classes (Informative)

Typical outputs include ToD, 1 PPS, 10 MHz/5 MHz, packetized time (PTP), and host-bus time (PTM). Electrical characteristics and limits **SHOULD** be published for each.

5.7.3 - Stability, Accuracy, Precision (Normative Reporting)

Qualities sought include adequate stability (ADEV/TDEV/MTIE), low phase noise, high accuracy, high precision, and fine resolution.
Numeric targets are intentionally unspecified; vendors **SHALL** report measurements using: - ADEV/TDEV versus tau

- Time/frequency offset to reference
- Timestamp granularity
- Physical synchronization extent and conditions - And any other measurement the manufacturer chooses to include

5.7.4 - Phase Noise and Time Jitter (Normative Reporting)

Periodic outputs (e.g., 10 MHz) **SHOULD** include PN spectrum vs. offset frequency.
Pulse outputs (e.g., PPS), **SHALL** specify RMS and peak-to-peak time jitter and measurement bandwidth in Hertz. These requirements **MAY** be conditioned on intended signal and its intended use.

5.7.5 - Holdover Performance (Normative)

Holdover performance characterizes the stability of the TimeCard when all primary synchronization references are disconnected or lost. - Vendors **SHALL** publish the maximum holdover error bounds versus elapsed time (e.g., drift in microseconds over 4, 12, and 24 hours). - Vendors **SHALL** publish the warm-up conditions required before the oscillator's holdover stability guarantees become valid, and the test temperature range the holdover specification assumes.
- Holdover requirements strictly apply to the continuous physical drift of the 1PPS output, assuming the TimeCard was previously locked to a perfect, zero-noise inbound reference prior to disconnection for some specified minimum period of time.
- Maximum Time Interval Error (MTIE) per ITU-T G.8260 (or G.810 App II.5) **SHALL** be used as the definitive mathematical holdover metric. Other auxiliary holdover metrics (such as frequency aging rate) MAY also be measured and documented to assist system integrators.

Informative Note: Implementers should be cautious when relying solely on generic telecommunications boundaries (such as certain relaxed profiles within ITU G.8262.1), as those bounds are often too loose for the nanosecond-class strictness required by modern distributed datacenters and AI clusters. P3335 TimeCards target significantly tighter phase-drift boundaries.

5.7.6 - Ensemble References (Normative)

Implementations **SHALL** support combining multiple references into one unified "Ensemble" reference.
Ensemble logic **MAY** provide source weights, health, and alarms via management telemetry.

5.7.7 - Large-Extent Synchronization (Informative)

For data-hall or campus deployments, the intent is to achieve a specified maximum end-to-end time error while implementing calibration as needed, and meeting cable/optical constraints as a function of the physical dimensions (in meters) of the deployment extent. Many independent vendors are involved, so this is a matter of overall system design, and not of TimeCard design per se.

5.7.8 - Time-Flow Narrative (Informative)

The logical flow of time through the TimeCard architecture generally follows these stages:

1. **Ingress:** The TimeCard receives zero or more external references (e.g., GNSS, PTP, or PPS).
2. **Selection:** The system evaluates the health, stability, and configured priority of the incoming references and selects the most optimal source using a defined policy (e.g., Best Master Clock Algorithm).
3. **Disciplining (PLL):** The selected reference is fed into a phase-locked loop (PLL) which gracefully **« define »** disciplines the TimeCard's internal local oscillator. This loop filters out short-term jitter from the reference, relying on the high short-term stability of the local oscillator to provide a clean signal.
4. **Timescale Generation:** The disciplined oscillator drives a hardware counter, generating a single, unified timescale implementing an official timescale such as TAI or UTC.

5. **Egress:** The unified timescale is published across all Outbound Interfaces simultaneously. This includes generating physical PPS edges, updating memory-mapped Time of Day registers, scaling frequency outputs (e.g., 10 MHz), and serving host PCIe PTM requests—all originating deterministically from the exact same hardware counter.

If the last ingress reference is lost, the PLL freezes its correction values and the TimeCard enters **Holdover**, keeping the unified timescale running based purely on the uncorrected drift behavior of the local oscillator function.

5.7.9 - Implementation Flexibility (Informative)

The "oscillator function" need not be or contain a discrete resonator. A nuclear frequency source may be used for applications requiring extremenly good performasnce. A DDS or similar digital source may suffice for cost-sensitive designs.
SWaP-C trade-offs are left to design and the market.

5.7.10 - Conformance and Interface Definitions (Normative Guidance)

Undefined interfaces **SHALL** normatively cite approved **« By who, and how? »** Interface Definition Documents (IDDs) for interoperability.

Conformance testing **SHOULD** cover: - ADEV/TDEV measurement methodology
- PN masks and spectra
- PPS alignment across all outputs
- PTM latency/asymmetry characterization
- Holdover MTIE verification
- Ensemble behavior under reference loss - Plus anything else the manufacturer chooses

---

5.8 - Documentation Requirements (Normative)

Manufacturers **SHALL** provide publicly available datasheets specifying at least the following: - Which architectural principles and constraints are implemented
- PLL/disciplining loop type and bandwidth
- All performance metrics defined in §7 (for instance stability, accuracy, PN/jitter, holdover, ensemble)
- Traceability data sufficient for analysis, or an explicit "traceability not supported" statement
- All optional or conditional features provided in the implementation

---

5.9 - Vendor Datasheet Checklist (Informative)

« Explain the purpose of this section »

Category	Required / Recommended	Example Contents
Electrical & Power	REQUIRED	Input rails, tolerance, current draw, sequencing
Mechanical	REQUIRED	Dimensions, connectors, faceplate layout
Thermal	REQUIRED	Operating/storage temp, airflow, warm-up load
Timing Performance	REQUIRED	ADEV/TDEV plots, PN spectra, holdover MTIE
Synchronization Inputs	REQUIRED	GNSS/PTP/NTP/WR interface specs

Category	Required / Recommended	Example Contents
Outputs	REQUIRED	PPS, 10 MHz, ToD specs, alignment accuracy
Management	RECOMMENDED	Interface type, telemetry fields, firmware update method
Compliance	REQUIRED	Safety, EMC, ESD, RoHS, CE/FCC marks
Reliability	RECOMMENDED	MTBF, service intervals, backup power retention
Documentation	REQUIRED	Quick start guide, LED legend, calibration method

---

5.10 - References (Normative)

- [G8620] ITU-T G.8260 *Definitions and terminology for synchronization in packet networks* (2015 or later).

- [IEEE 1139] *IEEE Standard Definitions of Physical Quantities for Fundamental Frequency and Time Metrology.*
- [IEEE 1193] *IEEE Guide for Measurement of Environmental Sensitivities of Standard Frequency Generators*
- [MTIE] Stefano Bregni, *Measurement of Maximum Time Interval Error for Telecommunications Clock Stability Characterization*, IEEE Transactions on Instrumentation and Measurement, Vol. 45, No. 5, Oct 1996.

- [PTPv2.1] IEEE 1588-2019 *Precision Time Protocol (PTP).*

5.11 - Bibliography (Informative)

- Wikipedia: [Precision Time Protocol] (https://en.wikipedia.org/wiki/Precision_Time_Protocol)

- [IRS_DID] DI-IPSC-81434A, *Interface Requirements Specification Data Item Description* (1999).

- [IDD_DID] DI-IPSC-81436A, *Interface Design Description Data Item Description* (1999).

- [SSDD] DI-IPSC-81432A, *System/Subsystem Design Description* (1999).

- [SSS] DI-IPSC-81431A, *System/Subsystem Specification* (2000).
- NIST Special Publication 1065 by Riley

5.12 - Notes

The present P3335 standard document was initiated on 25 April 2025, largely based on *"TimeCard Architecture (Section 5) Draft (20250424).docx"*.

5.13 - Acronyms

1PPS = One Pulse Per Second
ADEV = Allan Deviation
ASIC = Application Specific Integrated Circuit
DDS = Direct Digital Synthesis
EMC = Electro Magnetic Compatibility
ESD = Electro Static Discharge
FPGA = Field Programmable Gate Array
gRPC = Google Remote Procedure Call
GNSS = Global Navigation Satellite System
I2C = Inter-Integrated Circuit
IDD = Interface Definition Document
I/O = Input/Output
IRIG = Inter-Range Instrumentation Group
ISA = Industry Standard Architecture computer bus
ITU = International Telecommunications Union
LED = Light Emitting Diode
LPT = Line Printer Terminal
M&CI = Management and Control Interface
MCA = Micro Channel Architecture
MHz = Megahertz (10^6 Hertz)
MTBF = Mean Time Between Failure
MTIE = Maximum Time Interval Error
NTP = Network Time Protocol
PCIe = Peripheral Component Interconnect Express
PCMCIA = Personal Computer Memory Card International Association
PLL = Phase Locked Loop
PN = Phase Noise
PTM = Precision Time Measurement (Intel)
PTP = Precision Time Protocol
REST = Representational State Transfer
RMS = Root Mean Square
SCSI = Small Computer System Interface
SMB = System Management Bus
SNMP = Simple Network Management Protocol
SoC = System on a Chip
SWaP-C = Size, Weight, Power, and Cost
TAI = International Atomic Time (in French)
TDEV = Time Deviation
ToD = Time of Day
USB = Universal Serial Bus
UTC = Coordinated Universal Time
WG = Working Group
WiWi = Wireless two-Way interferometry
WR = White Rabbit
WWVB = Radio Station WWVB

End of Document

Performance Metrics (Normative)

This clause defines the performance reporting and characterization requirements for TimeCard implementations. The intent is to enable measurable, transparent, and comparable timing behavior across vendors. Performance metrics shall be reported in vendor documentation.

6.1 Overview

TimeCard performance must quantify the precision, stability, and accuracy with which the device maintains and delivers time and/or frequency in a context aligned with the performance requirements of the intended application.

Performance metrics for the oscillator(s) utilized within the TimeCard shall be provided. These shall include frequency accuracy and temperature stability.

Performance metrics applicable for the characterization of the operation within the applicable use case should be provided. Those metrics should be characterized under both locked (synchronized/syntonized) and holdover operating conditions.

Manufacturers may report the following metrics: - Frequency stability (ADEV, TDEV) - Maximum Time Interval Error (MTIE) - Phase noise (PN) - Jitter and alignment (PPS and ToD) - Accuracy and drift relative to reference - Holdover behavior - Environmental sensitivity (to temperature, voltage, vibration)

All measurements shall be made by testing equipment traceable to the relevant standards, such as UTC(NIST), and conform to relevant methodologies e.g. ITU-T G.810/G.8260.

6.2 Benchmarking and Testing Oscillators on the TimeCard

To validate the capabilities of the onboard oscillator(s) within the context of the TimeCard system, comprehensive testing procedures shall be conducted. The presence of the TimeCard inside a host system (such as a PCIe slot in a standard server) introduces environmental stressors such as thermal gradients, voltage fluctuations, and vibrations, which must be accounted for during benchmarking.

6.2.1 Test Environment and Setup

Benchmarking shall be performed using industry-standard precision measurement equipment to ensure reproducibility and traceability: - **Reference Sources:** An atomic reference standard (e.g., Cesium or Rubidium clock) or a highly stable GNSS-disciplined oscillator (GNSSDO) tracing back to UTC. - **Measurement Instruments:** High-resolution Time Interval Counters (TIC), Phase Noise Analyzers, and Oscillators with low intrinsic noise. - **Environmental Controls:** Thermal chambers and vibration tables for simulating realistic operation conditions.

6.2.2 Core Performance Characterization

The following tests are standard for evaluating the intrinsic performance of the oscillator on the TimeCard:

1. **Fractional Frequency Stability (ADEV, TDEV):**
   - Measured by comparing the TimeCard's oscillator output (e.g., 10 MHz or 1 PPS) against the reference source over various observation intervals (τ).
   - Allan Deviation (ADEV) and Time Deviation (TDEV) metrics shall be plotted to characterize short-term, medium-term, and long-term instability (white noise, flicker noise, and random walk).
2. **Phase Noise Testing:**
   - Evaluated in the frequency domain using a Phase Noise Analyzer to quantify short-term phase fluctuations.
   - Typically reported at standard offset frequencies (e.g., 1 Hz, 10 Hz, 100 Hz, 1 kHz, 10 kHz, 100 kHz) from the carrier.

3. **Maximum Time Interval Error (MTIE) and Time Variance (TVAR):**
    - Assesses the peak-to-peak variation in time delay of the oscillator's output relative to an ideal reference over a specified observation window.
    - Crucial for determining whether the oscillator's wander meets telecom and networking standards (e.g., ITU-T G.8262).

6.2.3 Operational and Holdover Benchmarking

The interaction between the synchronization subsystem and the oscillator must be tested under dynamic conditions.

1. **Lock Acquisition and Tracking:**
    - Measure the time required to achieve syntonization and synchronization from a cold start and a warm start.
    - Characterize phase transients and frequency overshoot during the locking phase.
2. **Holdover Performance:**
    - **Procedure:** Allow the TimeCard to reach a fully stabilized locked state using a GNSS or PTP reference. Disconnect the reference source to force the system into holdover mode.
    - **Measurement:** Record the time drift (phase error) accumulation relative to the reference over specific periods (e.g., 1 hour, 4 hours, 24 hours).
    - The reported holdover class must specify the maximum time error observed over a defined duration and temperature profile.

6.2.4 Environmental Stress Testing

Since TimeCards are intended for deployment in standard computing hardware, the oscillators must be benchmarked under simulated data center or edge environments.

1. **Thermal Stability Profile:**
    - Expose the TimeCard to a defined temperature ramp (e.g., -5°C to +55°C depending on chassis requirements).
    - For high-precision applications, determine the dynamic thermal gradient response (e.g., drift resulting from a 1°C/minute change) and steady-state frequency over temperature (Δf/f vs. ΔT).
2. **Power Supply and Voltage Variation:**
    - The PCIe bus provides power, but servers can exhibit significant voltage ripple.
    - Test frequency stability and phase noise under $\pm$5% or $\pm$10% input voltage fluctuations, verifying the effectiveness of onboard Low Dropout Regulators (LDOs) and power-filtering circuitry.
3. **Vibration and Airflow:**
    - Data center chassis fans cause mechanical vibration and rapid, localized temperature shifts due to airflow.
    - Benchmark the oscillator using a shaker table matching typical server rack vibration profiles (e.g., ETSI EN 300 019).
    - Measure vibration-induced phase noise and dynamic frequency shifts.

6.2.5 PCIe and System Integration Impact

It is critical to test the oscillator's performance when integrated within a standard PCIe environment, rather than isolated on a test bench: - Compare baseline (standalone test bench) oscillator performance to performance while executing heavy PCIe data transfers. - Verify that standard clock domains within the server do not induce crosstalk or correlated jitter on the TimeCard's precision timing signals.

7. Timing Interfaces (Normative)

This chapter defines the physical, electrical, and logical interfaces through which a TimeCard exchanges timing and frequency information with external systems. These interfaces form the foundation of interoperability between diverse TimeCards, host platforms, and external timing networks. All implementations SHALL adhere to the standardized behaviors defined herein to facilitate cross-vendor compatibility and predictable operation.

7.1 Overview

A TimeCard typically incorporates three fundamental classes of interfaces: 1. **Receive (Inbound) Interfaces:** Dedicated to acquiring timing signals from external authoritative references. 2. **Providing (Outbound) Interfaces:** Dedicated to distributing disciplined time, phase, and frequency representations to the host system and other downstream consumers. 3. **Management and Control Interfaces:** Non-timing-critical data paths used for configuration, telemetry, security, and diagnostics.

Each class of interface possesses distinct electrical, signaling, protocol, and performance requirements, which are detailed in the following subclauses.

7.2 Receive (Inbound) Interfaces

The receive interface enables the TimeCard to lock its internal oscillator function to one or more external references, establishing traceability and synchronization to a higher-order time source.

7.2.1 Supported Reference Types

The following external references are recognized for TimeCard synchronization. A TimeCard MAY support one or more of these inputs: - **GNSS Receivers:** (e.g., GPS, Galileo, GLONASS, BeiDou) supplying 1 Pulse Per Second (1PPS) and Time of Day (ToD) streams such as NMEA or binary protocols. - **Precision Time Protocol (PTP):** Packet-based synchronization via IEEE Std 1588 over Ethernet or PCIe networks. - **Network Time Protocol (NTP):** Packet-based synchronization for administrative or lower-accuracy time distribution via RFC 5905. - **White Rabbit (WR):** Sub-nanosecond synchronization via deterministic Ethernet variants. - **WiWi:** Wireless precision timing leveraging IEEE 802.1AS extensions. - **Regional Radio Standards:** (e.g., WWVB, DCF77, MSF) for localized low-frequency time distribution. - **Direct Electrical References:** 1PPS and continuous frequency (e.g., 10 MHz) continuous-wave or square-wave inputs for direct discrete phase locking or localized chaining.

7.2.2 Interface Characteristics

For each physical receive interface implemented, the manufacturer SHALL rigorously specify the following characteristics in the product documentation: - **Electrical level:** (e.g., 3.3V CMOS, LVTTL, RS-422, sinewave dBm). - **Input impedance and line termination:** (e.g., 50 Ω, 75 Ω, high-Z). - **Signal polarity and edge definition:** For pulse inputs, defining whether the rising or falling edge acts as the on-time phase marker.

Additionally, the lock acquisition time parameters and holdover entry/exit behaviors are configurable properties of the TimeCard and SHALL be documented.

7.2.3 Input Redundancy and Selection

TimeCards supporting multiple reference inputs SHALL implement source selection and failover logic (e.g., an implementation of the Best Master Clock Algorithm or a proprietary multi-source weighting ensemble).

The source priority, weighting, and failover policy SHALL be configurable via the management interface. During a reference transition (failover), the TimeCard internal oscillator SHOULD maintain phase continuity; however, derivative outputs are permitted minor bounded discontinuities as defined by the manufacturer's specification.

7.2.4 Sensitivity and Thresholds

The physical receive path SHALL tolerate nominal input amplitude variations of at least ±20% and reject high-frequency noise consistent with ITU-T G.8260 sensitivity definitions.

Loss-of-signal (LOS) detection boundaries and subsequent signal reacquisition thresholds SHALL be explicitly documented in the vendor's datasheet.

7.3 Providing (Outbound) Interfaces

The providing interface is responsible for delivering the unified, synchronized timescale generated by the TimeCard to the host platform or external physical devices.

7.3.1 Output Signal Classes

Typical providing interfaces include, but are not limited to: - **Pulse-Per-Second (PPS):** The primary second-aligned discrete reference pulse. - **Frequency Outputs:** Continuous clocks (e.g., 10 MHz, 5 MHz) used for frequency synchronization. - **Time of Day (ToD):** Serial-formatted absolute time messages (e.g., NMEA, IRIG-B, or proprietary binary telemetry). - **Packetized Time (PTP):** IEEE 1588 traffic delivered via Ethernet ports or directly into the host via PCIe networking abstractions. - **In-Bus Timing (PTM):** PCIe Precision Time Measurement transactions directly delivered to the host bridging architecture. - **Synchronous Clocks:** Discrete phase-aligned clocks for driving localized FPGA or SoC logic domains.

7.3.2 Electrical and Protocol Standards

All physical outputs SHALL comply with the following baseline signaling levels unless otherwise documented for specialized physical environments: - **PPS and frequency outputs:** 3.3 V CMOS or LVTTL utilizing 50 Ω source impedance. - **ToD / Serial outputs:** RS-232, RS-422, or RS-485 compliant signaling levels. - **PCIe / PTM interfaces:** Compliant with PCI-SIG 5.0 specifications (or later). - **Ethernet / PTP interfaces:** Compliant with IEEE Std 1588-2019 and/or IEEE Std 802.1AS-2020.

For each implemented providing interface, the manufacturer SHALL document the signal rise/fall times, maximum intrinsic jitter, source impedance, and absolute drive capability.

7.3.3 Synchronization Accuracy

All providing output signals SHALL align coherently to the single unified timescale generated natively by the TimeCard as described in the Architecture clauses.

For compliant high-precision designs, the 1PPS output alignment error SHALL NOT exceed 1 nanosecond RMS relative to the perfectly aligned internal master reference point. Output clocks SHOULD exhibit deterministic phase behavior and strictly bounded frequency deviations during any internal source failover event.

7.3.4 Hardware Timestamping

Providing interfaces SHALL support hardware-assisted timestamping where technically feasible within the protocol, including: - Direct physical PPS edge capture. - In-band PCIe PTM transaction stamping. - PTP transmit (egress) and receive (ingress) hardware packet timestamps.

Timestamping accuracy validation SHALL be performed using measurement equipment directly traceable to at least one recognized National Metrology Institute (e.g., NIST, NPL, or PTB).

7.4 Management and Control Interfaces

Management interfaces define the logical communications paths dedicated to device configuration, telemetry polling, and firmware lifecycle control. While these interfaces do not carry the critical synchronization timepath, they are indispensable for operational deployment.

7.4.1 Common Management Interfaces

Interface	Typical Use	Relative Bandwidth	Example Use Case
SMBus / I²C	Low-level baseboard configuration	Low	Host BMC telemetry, power sequencing, clock control
IPMI / NC-SI	Out-of-band server management	Medium	Platform health monitoring, remote chassis querying
PCIe Config Space	In-band direct host control	High	Runtime MMIO status registers, PCIe PTM enablement
Serial / USB	Maintenance and physical access	Medium	Localized firmware flashing, debug CLI access
REST / gRPC / SNMP	Remote network orchestration	Variable	Distributed datacenter timing management systems

7.4.2 Management Capabilities

Where a management interface is provided, it SHALL allow read and write access to at minimum the following operational vectors: - Active synchronization source selection and current fallback state. - Internal oscillator disciplining mode (e.g., locked, holdover, free-run, warming up). - Phase and frequency offset statistics relative to the reference. - Active alarm registers, event logs, and threshold warnings. - Firmware versioning, hardware revisions, and active security status.

7.4.3 Security and Sanitization Requirements

Given the critical nature of timing infrastructure, management interfaces SHALL adhere to strict security protocols: - All remote or networked management channels SHALL require standard authentication mechanisms (e.g., username/password, cryptographic tokens, or mutual TLS certificates). - Firmware payload updates SHALL mandate cryptographic signing and boot-time integrity verification to prevent unauthorized code execution. - Telemetry data crossing shared or external network boundaries SHOULD be encrypted. - **Data Sanitization:** TimeCards SHOULD support a cryptographic erase or sanitization procedure. Upon receiving a specific management command, or upon detection of a physical tamper event, the system SHALL permanently erase all sensitive internal state data, private keys, and operational logs, satisfying requirements for environments where the overall system must forget all internal data upon a sanitization procedure.

7.5 Physical and Logical Integration

7.5.1 Connectors and Labeling

Physical TimeCard front panels (brackets) SHALL label all timing and management ports clearly. For example: `[GNSS] [PPS OUT] [10MHz] [ToD] [MGMT] [USB]`

Manufacturers SHOULD adopt standardized color coding and iconography in accordance with Open Compute Project Time Appliance Project (OCP-TAP) guidelines where applicable.

7.5.2 Interoperability Goals and Requirements

- All providing interfaces on a single TimeCard SHALL publish the same mathematically aligned, unified timescale.
- Interfaces SHOULD support broad cross-vendor interoperability by utilizing only the common electrical and logical formats specified herein.
- **Architectural Objective:** While vendors MAY implement proprietary or optional extensions to the TimeCard interface, it is a primary design objective of this standard that such extensions do not natively disrupt or degrade baseline interoperability with standard host systems.

7.6 Summary

Timing interfaces construct the foundational boundaries of TimeCard interoperability, facilitating precise, deterministic, and hardware-agnostic synchronization. Compliance with the physical and logical interface definitions in this chapter realizes the goal that any TimeCard—regardless of underlying component design or vendor origin—can seamlessly integrate into diverse host environments, sustain traceable absolute time accuracy, and distribute stable frequency and phase alignments.

---

8. Control Interfaces (Normative)

This chapter defines the physical, logical, and protocol-level **control interfaces** used to configure, monitor, and manage TimeCard devices.

8.1 Overview

Control interfaces form the out-of-band and in-band **management plane** of the TimeCard architecture. Unlike the timing interfaces defined in Clause 7, which operate strictly within the data plane to deliver phase and frequency synchronization, control interfaces are responsible for: - Configuration of synchronization parameters and state machines. - Status monitoring, health telemetry, and fault reporting. - Firmware provisioning and lifecycle security management. - Traceability reporting and calibration operations.

Every compliant TimeCard implementation SHALL provide at least one accessible control interface. This interface MAY be exposed through a physical baseboard connector, a host-exposed bus interface, or a network-based protocol suite.

8.2 Control Interface Classes

Control interfaces are categorized based on their communication topology and data granularity.

Class	Description	Typical Use
Local Hardware Control	Direct access via SMBus, I²C, I3C, or GPIO.	Baseboard-level configuration, power sequencing, or low-level telemetry by a BMC.
Host-Integrated Control	In-band access via PCIe MMIO or USB.	Runtime synchronization control by host OS software or drivers.
Out-of-Band (OOB) Control	NC-SI, IPMI, or dedicated serial interface.	Independent platform management regardless of the host OS state.
Remote Network Control	REST, gRPC, Redfish, or SNMP.	Distributed system orchestration or cloud-scale management.

A TimeCard MAY implement one or more of these control classes simultaneously to support sophisticated datacenter abstraction models.

8.3 Minimum Functional Requirements

Where a control interface is implemented, it SHALL expose at least the following minimum common set of capabilities, restricted by the bandwidth and constraints of the specific medium:

Function	Description
Source Selection	Select, prioritize, or configure the active synchronization reference (e.g., GNSS, PTP, PPS).
Mode Control	Read and modify the current operational state machine (e.g., Locked, Holdover, Free-running, Warm-up).
Clock Discipline Settings	Adjust PLL or disciplining loop time constants and bandwidth parameters.
Telemetry Access	Retrieve metrics such as fractional frequency offset, internal temperatures, instantaneous phase error, and stability statistics.
Alarm and Event Logs	Report critical operational events such as loss of reference, spoofing detection, or firmware integrity faults.
Firmware Management	Support cryptographic firmware update triggers, version reporting, and rollback.

8.4 Communication Protocols

8.4.1 SMBus / I²C / I3C

- Provides low-speed, low-latency configuration and sensor access, typically used by a Baseboard Management Controller (BMC).
- The address space and register map SHOULD be modeled consistently across compliant implementations to ease integration.
- Bus arbitration, clock stretching, and timing compliance SHALL follow the respective SMBus or MIPI I3C specifications.

8.4.2 PCIe Configuration and MMIO

- Allows in-band, high-speed access to TimeCard control registers directly from the host CPU.
- The register layout SHOULD comply with recognized industry standard memory maps (such as the OCP-TAP Control Register Map) where applicable.
- If the TimeCard supports PCIe Precision Time Measurement (PTM), the control interface SHALL present the necessary standard capability structures to the host to enable PTM negotiation.

8.4.3 IPMI / NC-SI

- Enables robust out-of-band control independent of the host processor or operating system.
- TimeCard devices establishing IPMI communication SHOULD implement an extended command set for time synchronization management (e.g., querying clock state, initiating reference failover).
- IPMI commands MAY support authenticated sessions utilizing pre-shared platform credentials.

8.4.4 REST / gRPC / SNMP / Redfish

- Used for high-level, network-based management and large-scale telemetry aggregation.
- REST, Redfish, and gRPC APIs SHOULD conform to established Open Management Schemas (OMS).
- SNMP implementations SHOULD expose standard OIDs for timing health, synchronization source, and hardware/firmware versions.

8.4.5 Serial and USB

- Serial (RS-232/RS-422/RS-485) interfaces SHOULD provide a human-readable command-line shell or SCPI-like syntax for localized debugging.

- USB connections MAY present as standardized Communications Device Class (CDC) / virtual COM ports to facilitate maintenance or localized firmware provisioning.

8.5 Control Register Structure

The logical control registers listed below represent the fundamental baseline for low-level interaction between the host/BMC and the TimeCard. While physical implementations differ (e.g., I2C addresses vs. PCIe MMIO offsets), the logical concepts SHALL be present:

Logical Register Concept	Description	Access	Example Units
`SYNC_SRC`	Active external synchronization source ID	R/W	Enumeration
`PLL_STATE`	Current disciplining state (Locked, Holdover, Free)	R	Enumeration
`PHASE_ERR`	Instantaneous phase error relative to the primary reference	R	nanoseconds
`TEMP`	Active local oscillator temperature	R	°C
`MTIE_EST`	Current estimated holdover drift/MTIE boundary	R	nanoseconds
`FIRMWARE_VER`	Active running firmware version string	R	ASCII / Hex
`UPDATE_CMD`	Trigger flag for firmware update sequences	W	Boolean
`SECURE_MODE`	Runtime security lockdown indicator	R/W	Boolean

All registers SHOULD adhere to consistent endianness and word alignment across a vendor's product line.

8.6 Security and Access Control

Note: The requirements in this subclause apply conditionally. They are REQUIRED only for TimeCards marketed or designated for secure infrastructure deployments. For isolated or lab-grade physical environments, these features MAY be omitted to reduce complexity.

Where security hardening is implemented, control interfaces operate as primary attack surfaces and require robust cryptographic protection.

8.6.1 Authentication and Authorization

- Networked control protocols SHALL support user or machine authentication.
- Privilege levels SHOULD distinguish between read-only monitoring access, operator configuration access, and administrator provisioning access.
- Access credentials and cryptographic keys SHOULD be stored securely within a hardware-backed enclave or discrete TPM.

8.6.2 Secure Firmware and Configuration

- Firmware payload images SHALL be digitally signed using verifiable certificates.

- A secure boot sequence SHOULD verify firmware integrity (cryptographic hash matches) at initialization before granting control to the TimeCard operating logic.

8.6.3 Network Security

- Remote IP-based interfaces (REST, Redfish, gRPC) SHOULD utilize encrypted transport layers (e.g., TLS).
- Replay and injection protection SHOULD be implemented for operations traversing shared networking environments.

8.7 Event and Telemetry Management

To facilitate continuous performance observability, the control interface SHOULD support structured event and telemetry generation.

8.7.1 Event Reporting

- Generated operational events SHALL include a local timestamp, a severity classification, and a source identifier.
- Critical events SHOULD include, but are not limited to:
  - Reference loss (LOS)
  - Phase lock achieved or lost
  - Holdover entry or exit
  - Temperature threshold alarms
  - Firmware update success or failure.

8.7.2 Telemetry Streaming

- Continuous telemetry MAY be streamed via IPMI, REST, or gRPC for integration into centralized infrastructure dashboards.
- Telemetry generation rates SHALL be configurable by the administrator.
- Time synchronization statistics (such as ADEV or instantaneous phase offsets) SHOULD be strictly timestamped and aligned to the TimeCard's unified timescale.

8.7.3 Logging and Traceability

- Critical alarm logs SHOULD be preserved in non-volatile memory across power cycles.
- Exported log files MAY include cryptographic signatures to validate log integrity during forensic analysis.
- Traceability metadata (such as calibration constants and firmware build hashes) SHOULD be accessible to support operational compliance audits.

8.8 Interoperability and Extensions

All implemented control interfaces SHALL comply with the baseline schema defined by the overarching IEEE P3335 architecture.

Manufacturers MAY implement proprietary extensions, custom registers, or vendor-specific telemetry strings. However, as an architectural objective, these optional extensions SHOULD NOT compromise or interfere with the baseline interoperability of the standardized registers and capabilities.

8.9 Summary

Control interfaces construct the necessary framework for secure, consistent, and vendor-neutral lifecycle management of TimeCard systems.

By standardizing configuration and telemetry logic, this framework facilitates the integration of diverse TimeCards into heterogeneous server fleets and telecommunications infrastructures. This architectural consistency promotes long-term maintainability, traceability, and operational reliability throughout the lifespan of precision time synchronization networks.

End of Chapter – Control Interfaces

9. Environmental Specifications (Normative)

This chapter defines the environmental, mechanical, and reliability parameters applicable to compliant TimeCard implementations. The objective is to establish predictable hardware behavior under diverse deployment conditions—ranging from controlled data centers and telecommunications facilities to industrial or field installations—while maintaining timing precision and conformance to the unified timescale.

9.1 Overview

TimeCards are precision timing subsystems whose performance is directly influenced by environmental factors such as ambient temperature, thermal gradients, humidity, mechanical shock, vibration, and electromagnetic interference.

All implementations SHALL specify their environmental operating limits, test methodologies, and mitigation measures in compliance with the standards to which the manufacturer claims conformity.

Environmental performance SHOULD be validated through qualification testing to verify that frequency, phase, and absolute time accuracy remain stable and within stated bounds over the entire declared operating profile.

9.2 Operating Conditions

9.2.1 Temperature

- **Operating Range:** Each TimeCard SHALL explicitly define a continuous operating temperature range suited for its target deployment (e.g., 0°C to +55°C for data center use, or −40°C to +85°C for industrial use).

- **Storage Range:** Vendors SHALL specify safe, non-operational storage temperatures and humidity levels.

- **Thermal Stability:** Local oscillator frequency drift versus temperature (temperature coefficient) SHALL be characterized and documented.

- **Compensation:** Precision devices employing Oven-Controlled Crystal Oscillators (OCXO) or Temperature-Compensated Crystal Oscillators (TCXO) SHOULD document their integrated temperature compensation logic or calibration tables.

- **Thermal Alarms:** Active management telemetry, if present, SHALL include configurable over-temperature and under-temperature warning thresholds.

9.2.2 Humidity

- **Operating Humidity:** Devices SHALL be specified to operate safely between 5% to 95% non-condensing relative humidity, unless an alternative range is explicitly declared.

- **Environmental Sealing:** Conformal coating, potting, or physical sealing SHOULD be applied for models explicitly marketed for humid, corrosive, or outdoor environments.

- Condensation prevention bounds SHALL be documented, and survivability under thermal cycling SHOULD be verified.

9.2.3 Altitude and Airflow

- Operating altitude range and corresponding thermal derating curves SHALL be defined (e.g., typically 0–3000 meters above sea level).

- Host airflow assumptions SHALL be documented, including the minimum required airflow (e.g., Linear Feet per Minute - LFM or CFM) to guarantee safe heat dissipation as a direct function of the inlet air temperature.

9.3 Mechanical and Structural Requirements

9.3.1 Form Factor

- The physical envelope SHALL be documented and SHOULD conform to an industry-standard dimensioning specification (e.g., PCIe low-profile, full-height, OCP NIC 3.0, or standard MEZZ).

- Custom, proprietary, or deeply embedded modules SHALL explicitly document mounting hole patterns, insertion force limits, and recommended connector retention torque levels.

9.3.2 Shock and Vibration

- TimeCards SHOULD withstand mechanical shock and vibration profiles aligned with their intended use cases. *(Note: For example, standard server-grade profiles may utilize IEC 60068-2-6 and IEC 60068-2-27 for vibration and shock respectively, while aerospace or heavy industrial profiles may dictate far stricter constraints. A single shock requirement is not broadly applicable across all TimeCard deployment profiles).*

- The mechanical isolation of the oscillator from board-level vibration is a matter of proprietary design.

- The manufacturer SHALL document the maximum frequency stability degradation (e.g., acceleration sensitivity, Γ, measured in parts per g) under anticipated vibration levels.

9.3.3 Connectors and Retention

- RF and timing reference ports SHALL utilize locking, threaded, or high-retention physical connectors (e.g., SMA, SMB, MCX, MMCX).

- Faceplate physical labeling SHALL clearly identify the function of each exposed port (e.g., GNSS, PPS, 10 MHz, ToD, MGMT).

- Appropriate physical cable strain relief SHOULD be incorporated to prevent mechanical fatigue on critical RF solder joints.

9.4 Electrical and Power Environment

9.4.1 Power Supply

- TimeCards SHALL distinctly define input power rail nominal voltages and required absolute tolerances (e.g., +12 V ±5%, +3.3 V ±3%).

- Devices physically powered via standard host buses (e.g., PCIe slots) SHALL comply rigidly with the bus specifications for power sequencing, inrush current, and maximum continuous current draw limits.

- External power connectors SHOULD include localized reverse-polarity and surge protection logic.

- Localized energy storage (e.g., supercapacitors or lithium backup batteries) MAY be utilized to sustain RTC or holdover operation during brief power outages.

9.4.2 Electromagnetic Compatibility (EMC)

- Implementations SHALL meet the EMC emission and immunity requirements dictated by the destination market's regulatory bodies (e.g., EN 55032, EN 55035, FCC Part 15 Subpart B).

- Baseboard shielding, ground-plane flooding, and oscillator enclosure isolation to exclude EMI are matters of proprietary vendor design.

9.4.3 Electrostatic Discharge (ESD)

- ESD protection mechanisms SHALL be provided on all externally exposed connectors in accordance with relevant electrical standards (e.g., IEC 61000-4-2 limits for contact and air discharge).

- Safe handling procedures and static warning labels SHOULD be included in both manufacturing integration documentation and end-user manuals.

9.5 Environmental Qualification Tests

All TimeCards intended for commercial production SHOULD undergo a formalized qualification testing matrix. Vendors MAY utilize the following baseline parameters, or add validated procedures specific to their target market:

Test Profile	Reference Standard	Primary Purpose
Thermal Cycling	IEC 60068-2-14	Validate oscillator phase stability across temperature extremes.
Humidity Endurance	IEC 60068-2-78	Assess component corrosion limits and moisture protection.
Mechanical Shock	IEC 60068-2-27	Verify PCB fracture survivability and mechanical retention.
Random Vibration	IEC 60068-2-64	Characterize phase noise and frequency stability under active vibration.
ESD Immunity	IEC 61000-4-2	Verify localized protection against electrostatic discharge.
EMC Emission/Immunity	EN 55032 / EN 55035	Baseline regulatory compliance.
Power Interruption	IEC 61000-4-11	Evaluate state-machine restart and holdover recovery transitions.

Performance deviation limits recorded during post-qualification testing SHALL NOT exceed the vendor's formally declared tolerances.

9.6 Reliability and Lifetime

9.6.1 Mean Time Between Failures (MTBF)

- Vendors SHALL specify an expected MTBF utilizing recognized reliability modeling standard (e.g., Telcordia SR-332, MIL-HDBK-217F, or field-measured historical data).

- MTBF goals SHOULD be scaled to the target environment (e.g., >100,000 hours for standard datacenters, or >50,000 hours for harsh field conditions).

9.6.2 Wear-Out and Service Life

- Hardware components susceptible to finite lifespans or wear-out (e.g., lithium backup batteries, localized cooling fans, flash memory write cycles) SHALL distinctly list their rated service intervals in the product documentation.

- Essential internal device configuration and persistent calibration data SHALL safely persist across expected power cycles over the device's operational lifetime.

9.6.3 Calibration and Traceability

- TimeCards SHOULD undergo individual factory calibration against a traceable primary standard (e.g., UTC(NPL), UTC(NIST)) before delivery.

- Available calibration certificates SHOULD record the date of calibration, the strict environmental conditions present during the test, and the calculated margins of uncertainty.

Annex A (Informative): Supplemental Compliance and Documentation Guidelines

Editor's Note: The contents of this Annex are strictly informative. Hardware vendors are legally bound to comply with the regulatory constraints, safety laws, and environmental policies of the jurisdictions in which their products are manufactured and sold. This standard aims to aid interoperability, but conformity to this standard does not implicitly grant or enforce legal regulatory compliance.

A.1 Safety Standards

- TimeCard hardware is commonly evaluated against general electrical safety standards such as IEC 62368-1.
- Good engineering practices encourage the incorporation of over-voltage, over-current, and thermal shutdown thermal runaway protections.
- Required safety labels, high-voltage warnings, and chassis grounding instructions are typically made visible on the physical unit per market law.

A.2 Environmental Compliance

- Products sold into global markets generally conform to hazardous material directives such as RoHS, REACH, and WEEE.

- Hardware manufacturers are widely encouraged to utilize halogen-free PCB materials and design for extensive end-of-life recyclability.

A.3 End-of-Life Handling and Disposal

- It is recommended that vendors provide guidance for environmentally responsible hardware disposal.
- Best practices dictate that hazardous materials (e.g., lithium coin cells or backup batteries) are physically removed and sorted prior to e-waste recycling.
- A secure mechanism for the cryptographic erasure of firmware, private keys, and proprietary calibration data can significantly aid the secure decommissioning of units deployed in highly sensitive infrastructure.

A.4 Suggested Datasheet Inclusions

Comprehensive technical datasheets and integration manuals significantly reduce friction for end-users. Manufacturers are encouraged to document the following openly: - Complete environmental operating limits and a summary of qualification testing results. - Calculated MTBF and reliability datasets. - Proof of ESD and EMC compliance certifications. - Required power, thermal dissipation (BTU/hr), and CFM airflow requirements. - Publicly available API/Register mappings to support unhindered multi-vendor interoperability.

10. Applications and Best Practices (Informative)

This chapter provides informative guidance regarding key deployment scenarios, application domains, and operational best practices for integrating and maintaining TimeCard systems. It is intended to assist system architects, integrators, and facility operators in achieving optimal synchronization accuracy, stability, and reliability across diverse environments.

As an informative chapter, the guidelines presented herein are recommendations and do not constitute normative requirements for IEEE P3335 conformance.

10.1 Overview

TimeCards function as precision timing subsystems capable of providing highly stable, traceable, and interoperable time and frequency services to host platforms. Because the core architecture abstracts the complexities of hardware-level synchronization away from the host CPU, TimeCards are versatile enough to be deployed across a vast array of industries.

Adopting rigorous best practices for physical installation, software configuration, and continuous monitoring greatly increases the likelihood that a TimeCard will operate reliably at the peak of its specified performance envelope throughout its operational lifecycle.

10.2 Application Domains

The following subclauses detail the primary industries that leverage TimeCard architectures, noting the specific operational priorities and recommended configurations for each.

10.2.1 Data Centers and Cloud Infrastructure

- **Operational Purpose:** Coordinating distributed databases, globally synchronized transactions (e.g., Google Spanner/TrueTime equivalents), AI training cluster synchronization, and event logging across massive computing fleets.

- **Key Architectures:** PTP networks acting as the primary distribution method across the datacenter fabric, with TimeCards acting as Grandmaster clocks or high-precision Ordinary Clocks at the top-of-rack (ToR).
- **Recommended Best Practice:** Implement redundant TimeCards across multiple racks. Utilize PCIe Precision Time Measurement (PTM) to deterministically bridge the time from the network interface card directly to the host CPU domain, targeting sub-microsecond absolute accuracy.

10.2.2 Telecommunications and 5G/6G Networks

- **Operational Purpose:** Providing strict phase and frequency alignment for Open RAN (O-RAN) baseband units, fronthaul/backhaul cellular networks, and edge computing synchronization to prevent cellular interference.
- **Key Architectures:** Utilizing specialized telecom profiles such as ITU-T G.8275.1 (full timing support from the network) and G.8275.2 (partial timing support).
- **Recommended Best Practice:** Deploy a comprehensive ensemble of reference sources (e.g., GNSS augmented by network-delivered PTP). Operators are encouraged to actively monitor holdover Maximum Time Interval Error (MTIE) telemetry to anticipate network degradation during GNSS spoofing or jamming events.

10.2.3 Financial Systems and High-Frequency Trading (HFT)

- **Operational Purpose:** Enabling ultra-precise hardware timestamping for trading events to satisfy stringent regulatory compliance frameworks (e.g., MiFID II in Europe, SEC Rule 613 in the USA).
- **Key Architectures:** Direct 1PPS electrical signal distribution alongside high-frequency PTP broadcast networks, heavily reliant on hardware-timestamping at the exact point of ingress/egress.
- **Recommended Best Practice:** Maintain strict, mathematically provable traceability to Coordinated Universal Time (UTC) utilizing GNSS-disciplined master clocks. Calibration certificates SHOULD be maintained and refreshed annually to satisfy regulatory audits.

10.2.4 Power Grid and Industrial Control Systems (ICS)

- **Operational Purpose:** Synchronizing distributed control systems, phasor measurement units (PMUs), SCADA networks, and high-voltage protection relays to monitor wide-area grid stability.
- **Key Architectures:** Utilizing the IEEE C37.238 PTP Power Profile over ruggedized deterministic Ethernet.
- **Recommended Best Practice:** Focus on environmental hardening. TimeCards deployed in substations typically require extended operating temperature ranges (-40°C to $+85$°C) and robust electromagnetic interference (EMI) shielding to survive high-voltage switching transients.

10.2.5 Scientific Research and Metrology

- **Operational Purpose:** Distributing phase-coherent time signatures across particle accelerators, radio telescope arrays, and advanced metrology laboratories.
- **Key Architectures:** Leveraging White Rabbit (WR) optical networks or specialized continuous-wave RF distribution (e.g., 10 MHz sine waves) to achieve sub-nanosecond precision.
- **Recommended Best Practice:** Utilize high-stability local oscillators (such as Rubidium atomic clocks or high-end OCXOs). Operators are encouraged to continuously archive Allan Deviation (ADEV) and Time Deviation (TDEV) metrics to establish long-term baselines for experiment validation.

10.3 Deployment Best Practices

10.3.1 Hardware and Physical Installation

- **Thermal Management:** Precision oscillators are highly sensitive to thermal gradients. Installers are advised to verify that the host platform provides adequate, steady airflow across the TimeCard and to avoid placing the card adjacent to high-heat-dissipating components (e.g., flagship GPUs) un-baffled.
- **Cabling Quality:** For RF inputs and PPS outputs, operators are encouraged to utilize phase-stable, heavily shielded coaxial cables. Connectors (e.g., SMA) are best torqued to the manufacturer's exact specifications to maintain a consistent 50 Ω impedance boundary.
- **GNSS Antennas:** Route antenna cabling well away from high-noise switching power supplies or motorized equipment to minimize EMI ingress that degrades the signal-to-noise ratio.

10.3.2 Configuration of Synchronization Hierarchy

- **Source Prioritization:** Administrators are advised to configure a distinct reference hierarchy based on precision, stability, and availability (e.g., defining Local GNSS as Primary, Network PTP as Secondary, and adjacent 1PPS as Tertiary).
- **Ensemble Algorithms:** If the TimeCard supports multi-input ensemble modes, enabling this feature allows the card to mathematically weigh multiple high-quality references simultaneously, smoothing out transient anomalies from any single source.
- **Holdover Tuning:** Carefully tune holdover operational thresholds based on the characterized stability of the specific local oscillator model deployed, preventing the TimeCard from serving degraded time for too long after a reference loss.

10.4 Operational and Maintenance Best Practices

10.4.1 Monitoring, Telemetry, and Alarming

- **Continuous Observation:** Leverage out-of-band management interfaces (like IPMI, NC-SI, or dedicated I2C/SMBus bridges) to continuously poll synchronization health without burdening the host CPU.
- **Log Aggregation:** Stream frequency and phase offset telemetry into centralized time-series databases. This allows for the historical analysis of oscillator aging and network-induced jitter.
- **Threshold Alarms:** Configure proactive SNMP traps or Redfish events for critical state changes, such as reference loss (LOS), entry into holdover states, or sudden internal thermal spikes.

10.4.2 Firmware Lifecycle and Security

- **Cryptographic Verification:** Administrators are strongly advised to deploy only firmware packages that are cryptographically signed by the original hardware vendor.
- **Secure Staging:** Test all firmware updates on a staging TimeCard mirroring the production environment before executing fleet-wide deployments.
- **Access Control:** Restrict remote network access (e.g., REST/gRPC configuration ports) to isolated management VLANs, and routinely rotate authentication credentials.

10.4.3 Calibration and Traceability Maintenance

- **Routine Recalibration:** High-end OCXOs and atomic sources experience natural, slow frequency drift over years of operation (aging). Recalibrating the hardware against a primary standard (e.g., UTC(NIST)) minimizes this accumulated error.
- **Traceability Documentation:** Preserve and catalog the factory calibration certificates, as well as the logs of operational reference links, to simplify compliance audits in regulated industries.

10.5 Common Pitfalls and Mitigation Strategies

The following table outlines frequent deployment issues and recommended architectural or operational responses:

Issue / Symptom	Likely Root Cause	Recommended Mitigation Strategy
Constant 1PPS Misalignment	Reverse polarity settings, or significant cable-length propagation delay.	Verify rising-edge vs. falling-edge configurations. Calibrate for cable propagation delay (~5 ns per meter).
Intermittent Reference Loss	GNSS antenna sky-view obstruction, or localized RF jamming/spoofing.	Implement a secondary network-delivered PTP reference as a fallback. Employ anti-spoofing GNSS receivers.
Excessive Drift in Holdover	Drastic thermal fluctuations affecting the oscillator during the holdover period.	Improve host chassis thermal regulation. Utilize a TimeCard model equipped with a higher-grade OCXO.
High Host-to-Card Latency	Relying on software interrupts and unoptimized OS networking stacks to fetch time.	Transition to in-band hardware timestamping mechanisms such as PCIe PTM to bypass OS scheduling jitter.
Management Interface Timeout	Host CPU exhaustion preventing the OS from responding to in-band telemetry polls.	Shift telemetry polling to out-of-band (OOB) BMC channels like SMBus or NC-SI that operate independently of the host OS.

10.6 Summary

By implementing the engineering, integration, and operational best practices detailed in this chapter, system architects mitigate the risks associated with distributing highly precise time across complex distributed systems.

These informative recommendations complement the normative hardware and logical requirements defined within the broader IEEE P3335 specification. When applied comprehensively, they facilitate the deployment of robust, resilient, and highly traceable TimeCard infrastructures that perform reliably across decades of service.

Annex A: Metrics (Informative)

This annex provides definitions, mathematical measurement methodologies, and interpretation guidelines for the key timing and frequency metrics utilized throughout the IEEE P3335 TimeCard specification. Because this is an informative annex, the metrics and recommendations presented herein do not constitute strict normative requirements, but rather establish a common engineering vocabulary to facilitate consistent evaluation of stability, accuracy, and performance across different hardware implementations and test environments.

All metrics referenced in this annex heavily align with established industry metrology literature, specifically ITU-T Recommendations G.810 and G.8260, as well as IEEE Std 1139.

A.1 Overview

Metrology metrics quantify the operational performance of TimeCards across four primary domains: 1. **Time Stability:** The long-term coherence of the clock's phase. 2. **Frequency Stability:** The consistency of the clock's oscillator over specified averaging periods. 3. **Phase Noise:** The short-term, frequency-domain spectral purity of the oscillator. 4. **Holdover Behavior:** The predictable degradation of the timescale when external references are lost.

Utilizing a standardized mathematical framework allows operators to confidently compare devices from competing vendors and validates interoperability across heterogeneous networks.

A.2 Core Timing and Stability Metrics

A.2.1 Allan Deviation (ADEV)

Definition: A statistical measure characterizing the fractional frequency stability of an oscillator or a timing system as a function of the averaging time (τ). Unlike standard variance, ADEV converges for common oscillator noise types (such as flicker frequency noise).
Formula:

$$\sigma_y(\tau) = \sqrt{\frac{1}{2(N-1)} \sum_{i=1}^{N-1} (\bar{y}_{i+1} - \bar{y}_i)^2}$$

where $\bar{y}_i$ represents the continuous, successive fractional frequency averages measured over intervals of duration τ, and N represents the total number of frequency samples.

Purpose: ADEV is the primary metric for evaluating both short-term and long-term oscillator frequency stability.
Reference: ITU-T G.810 Annex I; IEEE Std 1139.

Typical Baselines for Reference:

Oscillator Technology	Typical ADEV ($\tau = 1$ s)	Typical Application
TCXO (Temperature-Compensated)	1×10^{-9}	Cost-sensitive edge compute
OCXO (Oven-Controlled)	1×10^{-11}	Datacenter / Telecom boundary clocks
Rubidium (Atomic)	1×10^{-12}	Primary Reference Clocks (PRC) / Core
CSAC (Chip-Scale Atomic)	5×10^{-12}	Portable / SWaP-constrained field units

A.2.2 Time Deviation (TDEV)

Definition: The time-domain equivalent of ADEV. It represents the measure of time stability related to the phase variations of the signal.
Formula:

$$\text{TDEV}(\tau) = \frac{\tau}{\sqrt{3}} \times \text{Modified ADEV}(\tau)$$

Purpose: TDEV specifically quantifies the variation in absolute time error over a given observation period.
Reference: ITU-T G.810 Annex I.
Usage: TDEV is highly useful for evaluating the precision of synchronization networks, particularly assessing packet delay variation (PDV) noise in PTP deployments.

A.2.3 Maximum Time Interval Error (MTIE)

Definition: The maximum peak-to-peak phase or time deviation observed between any two measurement points within an observation window of duration τ.
Formula:

$$\text{MTIE}(\tau) = \max_{i,j} |x_j - x_i|, \quad \text{for all } (t_j - t_i) \leq \tau$$

where x is the time error constraint.

Purpose: MTIE acts as a worst-case boundary metric, measuring the absolute maximum wander or drift of the timescale.
Reference: ITU-T G.8260 Appendix II.5.
Usage: MTIE is universally utilized to define strict holdover performance boundaries (e.g., assessing if a clock drifts more than 1.5 μs over a 24-hour holdover period).

A.3 Frequency Domain and Jitter Metrics

A.3.1 Phase Noise ($\mathcal{L}(f)$)

Definition: The power spectral density of phase fluctuations of a continuous periodic signal, traditionally expressed in decibels relative to the carrier per hertz (dBc/Hz) at varying offset frequencies from the main carrier frequency.
Formula:

$$\mathcal{L}(f) = 10 \log_{10} \left(\frac{S_\phi(f)}{2 \text{ rad}^2} \right)$$

where $S_\phi(f)$ is the one-sided spectral density of the phase deviations.

Purpose: Phase noise evaluates the short-term spectral purity of the local oscillator.
Reporting Profile: Manufacturers typically plot phase noise across logarithmic decades: 1 Hz, 10 Hz, 100 Hz, 1 kHz, 10 kHz, and 100 kHz offsets.
Reference: IEEE Std 1139.

A.3.2 Time Jitter

Definition: The short-term variance or instability of a time-domain signal (such as a 1PPS edge or a 10 MHz square wave) from its ideal, mathematically perfect periodic position.
Measurement Types: - Peak-to-Peak Jitter (J_{pk-pk}): The absolute maximum time difference between the earliest and latest occurrence of the signal edge over the sample set. - **RMS Jitter (J_{rms}):** The root-mean-square average of the jitter samples, representing the common standard deviation of the timing error.

Measurement Best Practice: Jitter characterization is typically captured using calibrated Time Interval Counters (TIC) operating over a statistically significant dataset (e.g., 1×10^6 samples) with picosecond-level native resolution.

A.4 Accuracy, Precision, and Granularity

To prevent ambiguity in TimeCard documentation, developers are encouraged to consistently separate the definitions of accuracy, precision, and hardware granularity.

Metric	Contextual Definition	Example Target Profile
Accuracy	The degree of conformance or offset of the measured time directly to an absolute primary reference scale (e.g., UTC).	$<$ 100 ns deviation from UTC(NIST) target
Precision	The internal repeatability or statistical variance of a measurement operation under completely identical conditions.	$<$ 5 ns 1σ variance over 1,000 consecutive PTM transactions
Resolution	The absolute smallest physically distinguishable change the circuit can detect or measure.	10 ps inherent TDC (Time-to-Digital Converter) physical resolution
Granularity	The minimal mathematical discrete step available in the reported digital timestamps.	1 ns bit-level granularity within an IEEE 1588 packet payload

---

A.5 Holdover Metrics

When a TimeCard suffers a total loss of its inbound external reference signals (e.g., GNSS jamming or fiber cut), the active disciplining loop opens, and the unit transitions to **holdover**.

A.5.1 Holdover Time Error

Definition: The continuously accumulating deviation of absolute time driven by the free-running fractional frequency offset of the internal oscillator during holdover.
Mathematical Concept:

$$\Delta x(t) = x_0 + \int_0^t \Delta y(\tau) d\tau + \frac{1}{2} D t^2 + \epsilon(t)$$

where x_0 is initial phase error, Δy is the normalized frequency offset, D is the linear frequency aging rate, and $\epsilon(t)$ represents internal random noise.

Practical Example: If a TimeCard enters holdover with an uncorrected frequency offset of 1.15×10^{-11}, the generated 1PPS signal will mechanically drift by approximately 1μs per day of holdover (≈ 86.4 ns per 1×10^{-12} offset).

A.5.2 Warm-Up and Recovery Behavior

- **Warm-Up Time:** The total duration required for an oscillator (specifically heated OCXOs or Rubidium cells) to reach thermal equilibrium and electrical stabilization upon a cold power cycle.
- **Recovery Time:** The architectural time required for the PLL to re-establish a solid lock and compress phase errors after successfully reacquiring a valid reference.
- **Note:** It is highly recommended that vendors empirically characterize and publish both parameters in standard hardware datasheets.

A.6 Ensemble and Correlation Metrics

In distributed environments, multiple TimeCards or multiple reference inputs may be mathematically combined into an **Ensemble Clock**.

Concept	Description
Ensemble Average Stability	Statistical improvement in composite ADEV proportional to $\sqrt{N}$, where N represents the number of completely independent, uncorrelated clocks operating in the ensemble.
Correlation Coefficient (ρ)	A statistical measurement quantifying the dependency or shared error vectors between clocks. A highly resilient ensemble relies on sources with low correlation ($\rho \approx 0$).
Weighting Function	The active algorithm determining the proportional contribution of each individual clock/reference step, dynamically shifting based on real-time MTIE performance or variance.

A.7 Environmental Influence Metrics

Because oscillators physically react to their environment, static laboratory metrics rarely reflect real-world holdover stability unless environmental parameters are factored in.

Parameter	Metric Definition	Impact
Temperature Coefficient	The variance in fractional frequency ($\Delta f/f$) per degree Celsius (Δ°C).	Directly dictates holdover drift if the chassis undergoes drastic thermal shifts (e.g., an AC failure).
Vibration Sensitivity (Γ)	The relative frequency change per g of mechanical acceleration.	Critical factor for systems deployed in heavily vibrating racks or industrial transit enclosures.
Aging Rate	The constant, predictable, unidirectional long-term frequency drift of the resonator over time (days/years).	Determines how frequently the TimeCard requires absolute external recalibration to prevent systemic clock bias.

A.8 Evaluating Hardware: Best Practices

When designing lab tests or automated qualification fixtures to evaluate the metrics detailed in this annex, engineers are encouraged to observe the following best practices: - **Traceability:** Utilize only highly calibrated measurement instruments (e.g., Phase Noise Analyzers, Counters) that possess verifiable traceability to primary national standards (NIST, PTB, NPL). - **Control Symmetries:** Rigorously design the physical measurement setup to minimize asymmetric cable propagation delays, thermal gradients, and RF impedance mismatches. - **Broad Spectrums:** Execute stability validation tests across broad averaging intervals ($\tau = 1$ s out to $> 10^5$ s) to capture both short-term jitter and diurnal temperature-driven wander. - **Contextual Logging:** Always log environmental metadata concurrently with performance data (e.g., ambient temperature, relative humidity, chassis airflow, and test equipment calibration schedules).

A.9 Summary

The mathematical metrics outlined in this annex establish a vital, unified vocabulary for characterizing the temporal performance of TimeCard architectures.

By standardizing exactly how stability, noise, drift, and holdover are mathematically calculated and reported, the IEEE P3335 ecosystem allows operators to rapidly compare discrete hardware configurations, validate regulatory compliance, and confidently integrate mixed-vendor timing infrastructure.

End of Annex A

Annex B: Test Procedures (Informative)

This annex provides standardized, repeatable engineering test procedures for evaluating the functional and performance characteristics of TimeCard implementations. These procedures are designed to ensure consistency, comparability, and mathematical traceability of measurement results across independent vendors, metrology laboratories, and live deployment environments.

As this annex is explicitly informative, the test methodologies proposed herein serve as best-practice engineering baselines for hardware validation rather than strict normative conformance requirements.

B.1 Overview

Comprehensive validation of a TimeCard architecture encompasses three primary testing domains: 1. **Functional Verification:** Validating logical compliance with the architectural interface, physical connectability, and management plane state-machine behaviors. 2. **Performance Validation:** Empirically measuring TimeCard frequency stability, phase noise, holdover drift, and timestamping jitter against the vendor's declared specifications. 3. **Environmental and Reliability Qualification:** Assessing physical performance consistency and survivability under externally applied, simulated stress conditions (e.g., thermal shock, vibration, or EMI).

To satisfy traceability requirements, it is strongly recommended that all tests be conducted utilizing equipment recently calibrated against recognized national metrology standards.

B.2 General Test Conditions

B.2.1 Environmental Setup

- Testing environments are typically maintained under nominal, thermally stable laboratory conditions unless a specific thermal extreme test is actively being executed. Baseline conditions generally sit at $23 \pm 3^{\circ}$C with 30–70% non-condensing relative humidity.
- The Device Under Test (DUT) should be allowed to reach complete thermal equilibrium and internal oscillator operational stability (warm-up) prior to the commencement of any metrology recording.
- Physical cabling (e.g., coaxial cables for 1PPS or 10 MHz) and RF splitters utilized in the test fixture should be phase-matched and carefully impedance-matched (typically $50\,\Omega$) to prevent signal reflections that cause false jitter readings in the instrumentation.

B.2.2 Power and Initialization

- Evaluators should apply host bus power (e.g., via a PCIe riser) or auxiliary power within the hardware's rated voltage tolerance.

- The initial power-up sequence provides a baseline to map the oscillator's raw warm-up time from cold-start to achieving a steady-state frequency lock.

B.2.3 Measurement Equipment Calibration limits

Instrumentation utilized in performance validation should possess a noise floor or error margin substantially lower than the DUT to ensure the DUT is being measured, rather than the noise of the instrumentation itself. - **Reference Clock:** The primary laboratory reference clock driving the test equipment should possess an Allan Deviation (ADEV) at least one order of magnitude ($10\times$) superior to the target specification of the DUT. - **Time Interval Counter (TIC):** TICs should possess a native single-shot resolution of < 10 ps for evaluating high-precision OCXOs or GNSS-disciplined TimeCards. - **Phase Noise Analyzer:** The analyzer's internal local oscillator should possess a phase noise floor at least 10 dB, preferably 15 dB, lower than the anticipated mask of the DUT across all relevant frequency offsets.

B.3 Functional Verification Tests

Functional tests validate the logical operations and state transitions of the TimeCard without necessarily grading sub-nanosecond physical precision.

B.3.1 Receive (Inbound) Interface Tests

1. **Reference Detection:** Sequentially apply valid external references (e.g., GNSS antenna input, PTP network feed, external 1PPS) and verify via the management telemetry that the TimeCard correctly recognizes the presence of the physical signal.
2. **Lock Acquisition:** Monitor the management plane to record the time taken for the TimeCard's internal Phase-Locked Loop (PLL) or disciplining mechanism to transition from "Free-Run" to "Locked" status. Compare this to the vendor's specified acquisition time limit.
3. **Failover Response:** While actively locked to a primary reference, abruptly disconnect the primary input. Verify that the hardware deterministically enters the "Holdover" state or smoothly transitions to a secondary reference without causing a systemic reboot or crashing the host driver.

B.3.2 Providing (Outbound) Interface Tests

1. **PPS Amplitude and Polarity:** Utilizing an oscilloscope bridged with a $50\,\Omega$ terminator, capture the 1PPS output edge. Verify the correct edge polarity (rising vs. falling), the slew rate (rise time), and the high/low voltage thresholds against standard CMOS/LVTTL specifications.
2. **PCIe PTM Validation:** If supported, execute host-side software to trigger Precision Time Measurement (PTM) dialogs across the PCIe bus. Correlate the returned PTM hardware timestamps against the physical 1PPS output to verify systemic, in-band latency and offset behavior.

B.4 Performance Validation Tests

Performance tests quantify the physical metrology capabilities of the TimeCard.

B.4.1 Frequency Stability (ADEV/TDEV)

- **Procedure:** Connect the TimeCard's primary continuous clock output (e.g., 10 MHz) to a Phase Noise Analyzer or high-resolution Frequency Counter. Record continuous phase/frequency samples locked against the laboratory primary reference.
- **Analysis:** Calculate and plot the Allan Deviation ($\sigma_y(\tau)$) and Time Deviation (TDEV) across logarithmic averaging intervals ($\tau = 1$ s, 10 s, 100 s, $1{,}000$ s, $10{,}000$ s).
- **Evaluation:** Compare the resulting stability curves against the manufacturer's published datasheet limits.

B.4.2 Short-Term Phase Noise ($\mathcal{L}(f)$)

- **Procedure:** Utilize a cross-correlated Phase Noise Analyzer to plot the Single Sideband (SSB) phase noise of the oscillator.
- **Analysis:** Measure the power spectral density in dBc/Hz at standard decade offsets extending from 1 Hz to 1 MHz from the primary carrier frequency.
- **Evaluation:** Validate that the plotted noise curve sits below the vendor's declared upper-bound phase noise mask.

B.4.3 Time Jitter

- **Procedure:** Capture a statistically significant sequence of 1PPS pulses (e.g., minimum 1×10^5 samples) utilizing a TIC triggered against the pristine laboratory 1PPS reference.
- **Analysis:** Calculate the Root-Mean-Square (RMS) jitter and the absolute peak-to-peak (J_{pk-pk}) timing variance of the signal edges.

B.4.4 Holdover Boundary (MTIE)

- **Procedure:** Allow the DUT to lock to a master reference and achieve thermal and mathematical equilibrium for at least 24 hours. Abruptly sever the reference connection, forcing the unit into holdover.
- **Analysis:** Continuously record the wandering time error of the TimeCard's 1PPS output against the laboratory master 1PPS for the specified holdover duration (e.g., 4, 12, or 24 hours).
- **Evaluation:** Compute the Maximum Time Interval Error (MTIE) across the observation window and verify the drift profile remains beneath the target limit (e.g., 1.5μs/24hr).

B.5 Environmental and Stress Qualification

Environmental tests validate that the TimeCard's hardware design robustly compensates for external physical interference.

Test Profile	Recommended Methodology	Verification Target
Thermal Cycling	Place the active, locked DUT inside an environmental chamber. Cycle the ambient temperature across the manufacturer's maximum specified operating range (e.g., 0°C to 55°C) using a predefined ramp rate (°C/min).	Assess the maximum frequency excursion during temperature gradients. Verify that the oscillator's internal temperature compensation mechanism functions correctly.
Power Interruption	Mechanically or electrically interrupt the host power plane for brief intervals (1 s to 10 s) if the TimeCard relies on localized holdover capacitors or batteries.	Verify that the hardware successfully bridges the interruption and maintains the internal unified timescale without experiencing a phase step or discontinuity upon power restoration.
Vibration / Shock	Subject the TimeCard to variable frequency, multi-axis harmonic vibration (5 Hz to 500 Hz) utilizing a specialized shaker table.	Record the immediate degradation in phase noise and frequency stability caused by g-force acceleration (Γ), verifying that the structural design adequately dampens the oscillator.

B.6 Reporting and Traceability

To maintain engineering transparency, comprehensive test reports for TimeCard validations should document: - A detailed block diagram describing the physical test fixture, including cable lengths to account for fixed propagation delays. - Expiration dates and calibration certificates for all primary laboratory measurement instrumentation used during the test. - The precise firmware revision, hardware board step, and baseboard management controller (BMC) version installed on the DUT. - Raw and plotted measurement data for all ADEV, TDEV, Phase Noise, and MTIE evaluations. - Explicitly stated conditions during holdover tests (e.g., the thermal ramp rate applied to the chassis while the reference was disconnected).

By adhering to a consistent, traceable testing methodology, integrators can confidently benchmark TimeCard implementations across diverse vendor ecosystems, guaranteeing reliable behavior in production infrastructure.

End of Annex B

Annex C: Bibliography (Informative)

This bibliography lists the key standards, engineering references, academic publications, and technical documents that provide foundational knowledge regarding the design, testing, and implementation of TimeCard architectures.

These documents are strictly non-normative. They are provided for informational purposes only, and their implementation is not strictly required to claim structural conformance to the IEEE P3335 standard.

C.1 International Standards and Industry Profiles

- [**B1**] CENELEC EN 55032 / EN 55035, *Electromagnetic compatibility of multimedia equipment – Emission and Immunity Requirements*, 2017.
- [**B2**] DMTF DSP0236, *Intelligent Platform Management Interface (IPMI) Specification v2.0*, 2015.
- [**B3**] IEC 60068-2-6, *Environmental testing – Part 2-6: Tests - Test Fc: Vibration (sinusoidal)*, 2007.
- [**B4**] IEC 60068-2-27, *Environmental testing – Part 2-27: Tests - Test Ea and guidance: Shock*, 2008.
- [**B5**] IEC 61000-4-2, *Electromagnetic compatibility (EMC) – Part 4-2: Testing and measurement techniques - Electrostatic discharge immunity test*, 2009.
- [**B6**] IEC 62368-1, *Audio/video, information and communication technology equipment – Part 1: Safety requirements*, 2018.
- [**B7**] IEEE Std C37.238™-2017, *IEEE Standard Profile for Use of IEEE 1588 Precision Time Protocol in Power System Applications*, 2017.
- [**B8**] IETF RFC 3411, *An Architecture for Describing Simple Network Management Protocol (SNMP) Management Frameworks*, 2002.
- [**B9**] ISO/IEC 17025, *General requirements for the competence of testing and calibration laboratories*, 2017.
- [**B10**] ITU-T Recommendation G.8275.1, *Precision time protocol telecom profile for phase/time synchronization with full timing support from the network*, 2016.
- [**B11**] ITU-T Recommendation G.8275.2, *Precision time protocol telecom profile for phase/time synchronization with partial timing support from the network*, 2017.
- [**B12**] PCI-SIG, *PCI Express Base Specification Revision 5.0 (incorporating Precision Time Measurement ECN)*, 2019.
- [**B13**] SMBus Forum, *System Management Bus (SMBus) Specification Version 3.2*, 2018.

C.2 Academic and Technical Metrology References

- [**B14**] Allan, D. W., *Time and Frequency (Time-Domain) Characterization, Estimation, and Prediction of Precision Clocks and Oscillators*, IEEE Transactions on Ultrasonics, Ferroelectrics, and Frequency Control, 1987.

- **[B15]** Bregni, S., *Measurement of Maximum Time Interval Error for Telecommunications Clock Stability Characterization*, IEEE Transactions on Instrumentation and Measurement, Vol. 45, No. 5, 1996.
- **[B16]** Calhoun, J., *Clock Stability Characterization for Network Synchronization*, IEEE Communications Surveys & Tutorials, 2009.
- **[B17]** CERN / GSI, *White Rabbit Specification and Documentation (Sub-nanosecond Ethernet synchronization)*.
- **[B18]** Department of Defense, *MIL-HDBK-217F: Reliability Prediction of Electronic Equipment*, 1991.
- **[B19]** Hartmann, J., *Open Timing Architectures for Data Centers*, OCP Time Appliances Project Whitepaper, 2020.
- **[B20]** Lombardi, M. A., *NIST Technical Note 1952: Introduction to PTP and synchronization techniques*, National Institute of Standards and Technology, 2016.
- **[B21]** Riley, W. J., *NIST Special Publication 1065: Handbook of Frequency Stability Analysis*, National Institute of Standards and Technology, 2008.
- **[B22]** Weiss, M., *Time Transfer and Synchronization for Distributed Systems*, NIST Technical Report, 2013.

C.3 Supporting Open Source and Community Resources

- **[B23]** Network Time Foundation, *NTP.org Reference Implementation (ntpd) source code and documentation.*
- **[B24]** Open Compute Project (OCP) Time Appliances Project (TAP), *TimeCard Hardware Schematics and Firmware Repository.* [https://github.com/opencomputeproject/Time-Appliance-Project]
- **[B25]** Open Compute Project (OCP) Time Appliances Project (TAP), *Interoperability Whitepaper and Multi-Vendor Testing Framework.*
- **[B26]** Turba Technologies, *Turba Analytics Documentation - AI workload and distributed timing analytics platform.*

C.4 Related Military/Defense Specifications (DID)

- **[B27]** DI-IPSC-81431A, *System/Subsystem Specification (SSS) Data Item Description*, 2000.
- **[B28]** DI-IPSC-81432A, *System/Subsystem Design Description (SSDD) Data Item Description*, 1999.
- **[B29]** DI-IPSC-81434A, *Interface Requirements Specification (IRS) Data Item Description*, 1999.
- **[B30]** DI-IPSC-81436A, *Interface Design Description (IDD) Data Item Description*, 1999.

End of Annex C